Stability thresholds of N-vortex rings on the hyperbolic plane (tanh model): rational Fourier blocks, palindromic threshold polynomials, and a certified table for $N \le 16$

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Abstract

Montaldi and Tokieda [1, §4.2] considered the image Green's function on the hyperbolic plane, whose pair interaction is $-\ln\tanh(d/2)$ (the Hamiltonian they attribute to Kimura). Boatto [2] computed the stability ranges of the regular N-ring in this model numerically. Here we treat the linear stability problem in closed form. In log-polar coordinates the Hessian of the augmented Hamiltonian at the ring splits into 2×2 Hermitian Fourier blocks whose entries are explicit rational functions of $\xi = \tanh^2(r/2)$, where r is the hyperbolic ring radius. Unlike the "sinh" model, the blocks have a radial–tangential coupling. Under $\xi \mapsto 1/\xi$ the diagonal entries are invariant and the coupling changes sign; this reciprocity comes from inversion in the boundary circle. It makes the threshold polynomials palindromic. For $N = 8, \dots, 16$ we certify the spectral threshold $\xi_*(N)$ (exact rational root isolation and Arb ball arithmetic) and the critical mode $\lfloor N/2 \rfloor$. We also certify, with PARI/GP and an independent mod-p factorisation test, that $\xi_*(N)$ has an irreducible palindromic minimal polynomial of degree $2N-2$, $\frac{3}{2}N-2$ or $\frac{3}{2}N-3$ according to N odd, $N \equiv 0$ or $N \equiv 2 \pmod 4$. The number $\xi_*(8)$ is an algebraic unit of degree 10; for $9 \le N \le 16$, $\xi_*(N)$ is not an algebraic integer. (That the thresholds are algebraic is automatic once the blocks are rational; the content is their explicit structure.) Rings with $N \le 7$ are stable at every radius. For odd $N \ge 9$ the energy–momentum (Hessian-definiteness) threshold lies strictly above the spectral threshold. In the gap the ring is linearly (spectrally) stable, and its nonlinear stability is undecided (a Krein window). In the variable $\tau = -\frac{N}{2}\ln\xi$ the blocks are, up to an m-independent term of order N^{-2}, a function of m/N. From this exact identity we prove, for every N and every radius, that $b_m > 0$ and that the critical mode is $\lfloor N/2 \rfloor$; for every even $N \ge 8$ and every odd $N \ge 17$ that the threshold is unique (for odd $9 \le N \le 15$ this follows from the certified computation); for every even N that it equals the energy–momentum threshold; and for every odd $N \ge 9$ that the Krein window is non-empty, of width $O(N^{-3})$. We also prove $\cosh r_* = \kappa_\infty N + O(1/N)$, with $\kappa_\infty = 1/(2\eta_\infty) \approx 0.1343122$ and η_∞ the root of $(\coth\eta - 1/\eta)^2 = 2\tanh(\eta)/\eta$. Physically, a large ring stabilises once neighbouring vortices are more than $\ell_* = \pi/\eta_\infty \approx 0.844$ curvature radii apart, which is exactly the critical spacing of an infinite vortex row on a horocycle. In the sinh model, $\cosh r_* \sim \sqrt{N}/2$.

1 Motivation and prior work

Setting. In §4.2 of [1] (arXiv:1210.5662v1, p. 25), for the Green's function $G = \log\frac{|z-w|^2}{|1+\lambda z\bar{w}|^2}$, Montaldi and Tokieda write: "On the hyperbolic plane $\lambda < 0$ this reduces to the Hamiltonian

adopted by Kimura [10]." After discussing $\lambda > 0$, they write:

> "The calculations based on this option of Green's function get so cumbersome that the stability problem seems no longer tractable analytically. It seems likely that the results are similar to those in section 3, though the details of where the bifurcations occur will differ."

At $\lambda = -1$ this Green's function is $\log \tanh^2(d/2)$, i.e. the model (1) below. This note treats the linear stability of the ring in this model. The Hessian at the ring reduces to explicit rational 2×2 blocks (Thm. 2). The threshold is the root of an explicit polynomial equation (§6), and for $N \leq 16$ it is certified together with its minimal polynomial (§7). Qualitatively the picture is that of their §2. Rings with $N \leq 7$ are stable at every radius; for $N \geq 8$ a ring is stable exactly beyond a threshold radius, and the critical mode is $\lfloor N/2 \rfloor$ (proved for every N; uniqueness of the threshold is proved for even N and odd $N \geq 17$ and certified for odd $N \leq 15$; §9, §7). There is one structural difference. The tanh blocks have a radial–tangential coupling, and for odd N it separates the spectral threshold from the energy–momentum threshold (§8); this has no counterpart in the sinh model. We do not study the bifurcations themselves, which are the subject of their §3.

Prior work on this model.

- Boatto [2] studies exactly this model: on the hyperbolic surface her Green's function is "$\Psi_K = -\frac{1}{2\pi} \ln \tanh(\frac{\sqrt{|K|}}{2} r)$". She writes the Hessian in the rotating frame in block form with "$\mathcal{Q}$ and $\mathcal{P}$ ... symmetric circulant matrices, while $\mathcal{G}$ is an asymmetric circulant matrix", i.e. the Fourier-block structure with a coupling term. She then "(numerically) monitor[s] the maximum and the minimum eigenvalues" and concludes that "for $N \geq 7$, we can only assure stability for sufficiently negative K" (her Fig. 5a, $N = 5, \ldots, 9$, ring of geodesic radius 1). So the stability question, the block structure and the qualitative picture for $N \geq 8$ are hers. Her criterion is Hessian definiteness, i.e. our energy–momentum threshold. Her ring has fixed geodesic radius 1 while K varies, so $K = -r^2$ with $\xi = \tanh^2(r/2)$. Implemented as written, her equations (14), (17) and (31) reproduce our thresholds exactly: $K_*(8) = -0.28218454857650$, i.e. $\xi_*(8)$ to 10^{-41}, and $K_{\mathrm{EM}}(9) = -0.48869837$, i.e. our $\xi_{\mathrm{EM}}(9)$, including the certified gap $\xi_{\mathrm{EM}} - \xi_* = 3.22 \cdot 10^{-8}$. Her Fourier blocks equal ours up to m-independent factors. Our implementation of her equations is in `tanh/boatto_check/`.
- Montaldi–Nava-Gaxiola [3] and Nava-Gaxiola's thesis [4] use the same Hamiltonian but analyse only $N = 2, 3$. Both point to Hwang–Kim [6] for rings.
- Hwang–Kim [6] investigate rings of vortices in detail; according to [4] they use the same Hamiltonian and give conditions for ring relative equilibria.

This note adds the closed-form blocks, the reciprocity, the certified table, the minimal polynomials and the odd-N window.

2 Scope and status labels

The companion note `sinh/note.tex` treats the sinh model, studied in [1]. We use these labels:

PROVED (all N)	Fourier-block theorem (Thm. 2); linearisation and stability criteria (Prop. 3); reciprocity (Prop. 5); threshold equation and even-N form. Exact symbol form of the blocks (Thm. 11); $b_m > 0$ and critical mode $\lfloor N/2 \rfloor$ for every N and radius (Lemmas 12, 13); a unique threshold for every even $N \geq 8$ (Prop. 15) and every odd $N \geq 17$ (Prop. 21); the large-N limit $\cosh r_* = \kappa_\infty N + O(1/N)$ and the critical spacing ℓ_* (Thm. 16); the horocyclic vortex row as the limit problem, stable iff its spacing is at least ℓ_* (Prop. 19, Cor. 20); energy–momentum threshold equal to the spectral one for every even N; a unique threshold for every $N \geq 8$ (Prop. 21, which combines the proofs for even N and odd $N \geq 17$ with the CERTIFIED Thm. 7(b) for odd $9 \leq N \leq 15$); a non-empty Krein window of width $O(N^{-3})$ for every odd $N \geq 9$, and the same limit κ_∞ (Thm. 18).
CERTIFIED (exact arithmetic, $N \leq 16$)	$b_m > 0$ on $(0, 1)$; critical mode $\lfloor N/2 \rfloor$; the value of $\xi_*(N)$ and the sign change of a_{crit} at ξ_*; $N \leq 7$ stable for all r; odd-N energy–momentum threshold strictly above ξ_*, with the critical block the only indefinite block on the window. Exact rational root isolation (sympy) and Arb balls (python-flint), §7.
CERTIFIED by computer algebra (PARI/GP), $8 \leq N \leq 16$	Irreducibility over $\mathbb{Q}$ of the threshold polynomial f_N, hence: f_N is the minimal polynomial of $\xi_*(N)$, the degrees (4) and leading coefficients of Table 1, palindromy of the minimal polynomial, $\xi_*(8)$ a unit, $\xi_*(N)$ not an algebraic integer for $N \geq 9$ (§7.1). Irreducibility is also checked in Lean (see below); the consequences (minimal polynomial, degree, not an algebraic integer) are drawn from it on paper.

Lean (checked in the repository's CI build)	In `lean/Havelock/TanhAlgebra.lean`, where the block formulas of Thm. 2 are definitions (their derivation from the Hamiltonian is item (vi)), Lean checks: (i) reciprocity of a_m and the even-N threshold equation; (ii) for every $2 \le N \le 16$, the statements of Thm. 7 on positivity, sign change and critical mode, with ξ_* located to within 10^{-20} in a rational interval; (iii) $\xi_*(8)$ is an algebraic integer and a unit; (iv) the odd-N Krein window is non-empty for $N = 9, 11, 13, 15$. (v) in `lean/Havelock/TanhLargeN.lean`, every PROVED statement of §9 (Thms 11–18, Prop. 19, Cor. 20, Prop. 21), including the rates and the row's per-pair Hessian entries; (vi) in `TanhBlocks.lean`, Lemma 1 and Thm. 2 derived from the Hamiltonian (criticality, the Hessian as circulant matrices, its Fourier blocks); (vii) in `LinearFlow.lean`, that the spectrum of the full $2N$-dimensional linearisation $\kappa_0^{-1}JH$ is the union of the block spectra, so that Prop. 3's criteria are statements about the whole system; (viii) in `TanhIrred.lean`, irreducibility of $f_8, \dots, f_{16}$ over $\mathbb{Z}$ and $\mathbb{Q}$ by mod-p certificates checked in the kernel (the minimal-polynomial, degree and non-integrality consequences are on paper, §7.1). Lean does *not* check that $\kappa_0^{-1}JH$ is the derivative of the vortex vector field (it is defined so), or nonlinear (Lyapunov) stability.
CONJECTURE	the degree pattern (4) and irreducibility/palindromy for all N.
NUMERICAL	The $1/N$ coefficients C_1 (derived, remainder not bounded); the constant W in the Krein-window width, and the critical block being the only indefinite block on the whole window for odd $N \ge 17$ (§9); the stability boundary of rows on hypercycles: the criterion $F(\pi) \ge 0$, the critical curvature $1/\sqrt{2}$ and an explicit sufficient spacing are DERIVED for the tanh model, that each κ has exactly one threshold is CERTIFIED (Arb), and the threshold values $\ell^*(\kappa)$ are computed (end of §9).

Nonlinear (Lyapunov) stability is claimed only in the form "positive-definite Hessian on the non-rotational modes". The passage from this to Lyapunov stability holds *modulo* $SO(2,1)$; see Remark 4. We have not checked the hypotheses of an energy–momentum theorem for this model against a source.

3 The model

3.1 Hamiltonian

We use the Poincaré disk $\mathbb{D} = \{|z| < 1\}$ with the metric $4|dz|^2/(1-|z|^2)^2$ and the area (symplectic) form $\omega = \sum_k \lambda(z_k)\, dx_k \wedge dy_k$, $\lambda(z) = 4/(1-|z|^2)^2$. The geodesic distance satisfies $\tanh(d(z,w)/2) = |z-w|/|1-z\bar{w}|$. For N unit vortices, take

$$H = -\sum_{j<k} \ln \tanh \frac{d_{jk}}{2} = \sum_{j<k} \Big(-\ln|z_j - z_k| + \ln|1 - z_j \bar{z}_k| \Big), \qquad J = \sum_k \frac{1+|z_k|^2}{1-|z_k|^2} = \sum_k \cosh r_k. \quad (1)$$

Here J generates rotations about 0, and Hamilton's equations are $\dot{x}_k = \lambda^{-1}\partial_{y_k}H$, $\dot{y}_k = -\lambda^{-1}\partial_{x_k}H$, as in `note.tex`. The function $f(d) = \ln\tanh(d/2)$ satisfies $f'' + \coth d\, f' = 0$ for $d > 0$ (since $f' = 1/\sinh d$). So $G(d) = -\frac{1}{2\pi}\ln\tanh(d/2)$ is the Green's function of $\Delta_{\mathbb{H}^2}$ that vanishes as $d \to \infty$.

Attribution. Montaldi–Tokieda [1, §4.2] state that the image Green's function $\log|z-w|^2/|1+\lambda z\bar{w}|^2$ "on the hyperbolic plane $\lambda < 0$ … reduces to the Hamiltonian adopted by Kimura". Montaldi–Nava-Gaxiola [3, eq. (12)] write Kimura's Hamiltonian as $-\frac{1}{4\pi}\sum\Gamma_i\Gamma_j\ln\frac{\langle X_i,X_j\rangle+1}{\langle X_i,X_j\rangle-1}$ with $\langle X,Y\rangle = -\cosh d$, which equals $-\frac{1}{4\pi}\sum\ln\tanh^2(d/2)$. Boatto [2] uses the same Green's function on the hyperbolic surface. We refer to (1) as the model attributed to Kimura [5].

No Robin term (intrinsic remark). $\mathbb{H}^2$ is homogeneous, so the regular part of G at the diagonal, measured in the intrinsic metric, is the constant $\lim_{d\to0}(G(d) + \frac{1}{2\pi}\ln d) = \frac{1}{2\pi}\ln 2$. The Robin (self-interaction) function is therefore constant, and (1) has no one-body self-term. This contrasts with the flat disk with a hard wall (§10), where the Robin function $\frac{1}{2\pi}\ln(1-|z|^2)$ is not constant and, combined with the Euclidean area form, changes the dynamics. In conformal coordinates, a one-body term $-\frac{1}{2}\sum\ln(1-|z_k|^2)$ does appear in (2) below. That term comes from the $j \neq k$ image sums, not from a self-interaction.

Relation to the sinh model. Since $\sinh(d/2) = \tanh(d/2)\cosh(d/2)$, $H_{\sinh} = -\sum\ln\sinh(d_{jk}/2) = H - \sum_{j<k}\ln\cosh(d_{jk}/2)$. A direct computation gives $\Delta_{\mathbb{H}^2}\ln\cosh(d/2) = \frac{1}{2}$. So the sinh model is the tanh model plus a uniform compensating background vorticity attached to each vortex. This is the continuation to $\mathbb{H}^2$ of the sphere model, where Gauss's law forces such a term. The sinh model is the $\lambda < 0$ member of Montaldi–Tokieda's family [1, (1.12)].

3.2 Power-sum decomposition

Let $P_p = \sum_j z_j^p$. Using $\ln|1-u| = -\operatorname{Re}\sum_{p\geq1}u^p/p$ and $\sum_{j\neq k}(z_j\bar{z}_k)^p = |P_p|^2 - \sum_j|z_j|^{2p}$, we get

$$H = -\sum_{j<k}\ln|z_j - z_k| \;-\; \frac{1}{2}\sum_j\ln(1-|z_j|^2) \;-\; \frac{1}{2}\sum_{p\geq1}\frac{|P_p|^2}{p}. \tag{2}$$

The series converges absolutely on $\mathbb{D}^N$. The last term $\Phi := -\frac{1}{2}\sum_p|P_p|^2/p$ is the source of the apparent intractability. On a ring it is diagonal in the Fourier modes.

4 Fourier blocks

4.1 Coordinates and the relative equilibrium

Let $w = e^{2\pi i/N}$ and write $z_j = \sqrt{\xi}\,w^j e^{s_j+i\theta_j}$, so the ring of Euclidean radius $\sqrt{\xi}$ is $s = \theta = 0$. Then $\omega = \sum_j\kappa(s_j)\,ds_j\wedge d\theta_j$ with $\kappa(s) = 4\xi e^{2s}/(1-\xi e^{2s})^2$, and $\kappa_0 := \kappa(0) = 4\xi/(1-\xi)^2$. Set

$$c := \cosh r = \frac{1+\xi}{1-\xi}, \qquad X := \xi^N, \qquad W_k := \sum_{\substack{p\geq1\\ p\equiv k\ (N)}} p\,\xi^p = \frac{k\xi^k}{1-X} + \frac{N\xi^{k+N}}{(1-X)^2} \quad (1\leq k\leq N).$$

In particular $W_N = NX/(1-X)^2$. Let $F = H - \Omega J$.

Lemma 1. *The ring is a relative equilibrium for every $\xi \in (0,1)$, with $\Omega\kappa_0 = -\dfrac{N-1}{2} + \dfrac{\xi}{1-\xi} - \dfrac{NX}{1-X}$. At the ring, $dF = 0$ in all $2N$ variables.*

Proof. All θ-derivatives vanish by the dihedral symmetry. For s_j: the pair term contributes $-\frac{N-1}{2}$ (its linear part is $-\frac{1}{2}\sum_{j<k}(s_j+s_k)$; see `note.tex`, §2). The one-body part is $V(s) = -\frac{1}{2}\ln(1-u) - \Omega\frac{1+u}{1-u}$ with $u = \xi e^{2s}$, and $V'(0) = \frac{\xi}{1-\xi} - \Omega\kappa_0$. For Φ, write $|P_p|^2 = \xi^p \sum_{j,k} w^{(j-k)p} e^{p\epsilon_j + p\bar{\epsilon}_k}$ with $\epsilon_j = s_j + i\theta_j$. Then $\partial_{s_j}|P_p|^2\big|_0 = 2p\,\xi^p \operatorname{Re}\sum_k w^{(j-k)p} = 2pN\xi^p[N \mid p]$, so $\partial_{s_j}\Phi|_0 = -N\sum_{q\geq 1} X^q = -NX/(1-X)$. Setting the sum to zero gives Ω. □

4.2 The block theorem

Theorem 2 (Fourier blocks; PROVED)**.** *In the coordinates (s,θ), the Hessian of F at the ring is block-circulant. On the Fourier mode $(s_j,\theta_j) = (S,T)\,w^{mj}$, $1 \leq m \leq N-1$, it acts as the Hermitian matrix*

$$M_m = \begin{pmatrix} a_m & ig_m \\ -ig_m & b_m \end{pmatrix},$$

where

$$\begin{aligned} a_m &= (N-1)c - \tfrac{1}{2}(c-1)^2 - \tfrac{m(N-m)}{2} + \frac{2NcX}{1-X} - \frac{N^2X}{(1-X)^2} \\ &\quad - \frac{N}{2}\,\frac{m\xi^m + (N-m)\xi^{N-m} + (N-m)\xi^{N+m} + m\xi^{2N-m}}{(1-X)^2}, \\ b_m &= \tfrac{m(N-m)}{2} + NW_N - \tfrac{N}{2}\,(W_m + W_{N-m}), \qquad g_m = \tfrac{N}{2}\,(W_m - W_{N-m}). \end{aligned}$$

On mode 0, $b_0 = g_0 = 0$. All entries are rational functions of ξ with rational coefficients. Moreover $a_{N-m} = a_m$, $b_{N-m} = b_m$, $g_{N-m} = -g_m$, and $g_{N/2} = 0$ for even N.

For comparison, the sinh model has $a_m = (N-1)(1+c^2)/2 - m(N-m)/2$, $b_m = m(N-m)/2$ and $g_m = 0$ (`note.tex`, Prop. 2).

Proof. Since $dF = 0$ at the ring (Lemma 1), the Hessian is well defined. We add the three contributions in (2) and the term $-\Omega J$.

Pair term. As in `note.tex` (Lemma 1 and the expansion before it), $-\sum_{j<k}\ln|z_j - z_k|$ has Hessian $-\frac{m(N-m)}{2}$ in ss and $+\frac{m(N-m)}{2}$ in $\theta\theta$ on mode m, with no $s\theta$ coupling.

One-body terms. $V(s)$ from the proof of Lemma 1 does not depend on θ. Using $\frac{d}{ds} = 2u\frac{d}{du}$,

$$V''(0) = \frac{2\xi}{(1-\xi)^2} - \frac{8\Omega\xi(1+\xi)}{(1-\xi)^3} = \frac{2\xi}{(1-\xi)^2} - 2c\,\Omega\kappa_0.$$

Substituting $\Omega\kappa_0$ and using $c - 1 = 2\xi/(1-\xi)$, so that $\frac{2\xi}{(1-\xi)^2} - \frac{2c\xi}{1-\xi} = -\frac{1}{2}(c-1)^2$, gives $V''(0) = (N-1)c - \frac{1}{2}(c-1)^2 + \frac{2NcX}{1-X}$. This term is the same on every mode and lies in the ss entry.

Power sums. The second-order part of $e^{p\epsilon_j + p\bar{\epsilon}_k}$ is $\frac{p^2}{2}(\epsilon_j^2 + 2\epsilon_j\bar{\epsilon}_k + \bar{\epsilon}_k^2)$. Summing with the weight $w^{(j-k)p}$: (i) the squares give $p^2N[N \mid p]\sum_j \operatorname{Re}\epsilon_j^2 = p^2N[N \mid p]\sum_j(s_j^2 - \theta_j^2)$; (ii) the cross term gives $p^2\big|\sum_j w^{jp}\epsilon_j\big|^2$. Hence the quadratic part of Φ is

$$\Phi_2 = -\frac{N}{2}W_N\sum_j(s_j^2 - \theta_j^2) \;-\; \frac{1}{2}\sum_{p\geq 1} p\,\xi^p\Big|\sum_j w^{jp}(s_j + i\theta_j)\Big|^2.$$

The first piece contributes $-NW_N$ to ss and $+NW_N$ to $\theta\theta$. For the second, expand $|\sum_j w^{jp}\epsilon_j|^2$ as a real quadratic form. It equals $\sum_{j,k}\cos\phi_{jk}(s_js_k+\theta_j\theta_k)+2\sum_{j,k}\sin\phi_{jk}\,s_j\theta_k$ with $\phi_{jk}=2\pi(j-k)p/N$. Its Hessian has circulant blocks $2\cos\phi_{jk}$ $(ss,\,\theta\theta)$ and $2\sin\phi_{jk}$ $(s\theta)$. On the vector w^{mk} their symbols are $N([p\equiv m]+[p\equiv -m])$ and $-iN([p\equiv m]-[p\equiv -m])$, using $\sum_k w^{(j-k)p}w^{mk}=Nw^{mj}[p\equiv m]$. Multiplying by $-\frac{1}{2}p\xi^p$ and summing over p gives $-\frac{N}{2}(W_m+W_{N-m})$ on both diagonal entries and $+\frac{iN}{2}(W_m-W_{N-m})=ig_m$ in the $s\theta$ entry.

Adding the three pieces gives a_m, b_m and g_m. In particular, $a_m=V''(0)-\frac{m(N-m)}{2}-NW_N-\frac{N}{2}(W_m+W_{N-m})$, and expanding W_m+W_{N-m} over the common denominator $(1-X)^2$ gives the bracket in the statement. For $m=0$, W_0 should be read as W_N, which gives $b_0=NW_N-NW_N=0$. The symmetries follow from $m\leftrightarrow N-m$. □

Checks. The blocks equal the Fourier symbol of the exact Cartesian Hessian of F (sympy second derivatives, with Ω solved numerically) to $\le 2\cdot 10^{-48}$. The grid was $N\in\{3,5,7,8,9,10,12,13,16\}$ and $\xi\in\{0.02,0.09,0.3,0.75\}$ (`check_jacobian.out`). An independent Cartesian Hessian at 60–110 digits, written without these formulas, matches the blocks to $2.5\cdot 10^{-59}$ (`review/compare_blocks.py`). The compact form of a_m agrees exactly with the block formula for $N=3..16$ and all m (`identities.out`).

5 Linearisation and stability criteria

Proposition 3 (PROVED)**.** *On mode m the linearised flow is $\kappa_0(\dot S,\dot T)^\top=\left(\begin{smallmatrix}0&1\\-1&0\end{smallmatrix}\right)M_m(S,T)^\top$. Its eigenvalues are*

$$\kappa_0\sigma=-ig_m\pm\sqrt{-a_mb_m}.$$

Hence the ring is spectrally stable iff $a_mb_m\ge 0$ for all $1\le m\le N-1$. The Hessian of F is positive definite on the non-rotational modes $1\le m\le N-1$ iff $a_m>0$ and $a_mb_m-g_m^2>0$ for all such m.

Proof. $\det\left(\left(\begin{smallmatrix}0&1\\-1&0\end{smallmatrix}\right)M_m-\lambda\right)=(\lambda+ig_m)^2+a_mb_m$. A 2×2 Hermitian matrix is positive definite iff its $(1,1)$ entry and its determinant $a_mb_m-g_m^2$ are positive. Mode 0 (rotation, and change of ξ along the family) carries $b_0=0$ and is excluded. □

So the coupling g_m only shifts the frequencies; it does not affect spectral stability. It does affect definiteness, and this is the origin of the odd-N window in §8.

Remark 4 (energy–momentum, "modulo $SO(2,1)$"). The variation dJ annihilates every mode $m\neq 0$. Hence $\ker dJ\subset(\text{modes } 1..N-1)\oplus\mathbb{R}\,\partial_\theta^{(0)}$, and positive definiteness on modes $1..N-1$ implies definiteness of d^2F on $\ker dJ$ modulo the rotation orbit. The full symmetry group of (1) is the isometry group $PSL(2,\mathbb{R})\cong SO(2,1)_0$. Its momentum map has three components, and the modes $m=\pm 1$ are tangent to the non-compact (boost) directions. We therefore state any Lyapunov conclusion only *modulo* $SO(2,1)$: up to motion along the group orbit. We do not claim stability modulo $SO(2)$ alone. We have not verified the hypotheses of a specific energy–momentum theorem (cf. [1, Thm. 2]) for the tanh model.

6 Reciprocity and the threshold equation

Proposition 5 (reciprocity; PROVED for all N,m)**.** *As rational functions of ξ, under the map $\xi\mapsto 1/\xi$,*

$$a_m(1/\xi)=a_m(\xi),\qquad b_m(1/\xi)=b_m(\xi),\qquad g_m(1/\xi)=-g_m(\xi).$$

Hence $M_m(1/\xi)=\overline{M_m(\xi)}$, and a_mb_m and $a_mb_m-g_m^2$ are invariant.

Proof. For $1 \le k \le N-1$ the closed form of W_k gives $W_k(1/\xi) = \xi^{N-k}\frac{(N-k)+kX}{(1-X)^2} = W_{N-k}(\xi)$, and $W_N(1/\xi) = W_N(\xi)$ since $X/(1-X)^2$ is invariant under $X \mapsto 1/X$. So $W_m + W_{N-m}$ is invariant and $W_m - W_{N-m}$ changes sign, which gives the claims for b_m and g_m. For a_m: under $\xi \mapsto 1/\xi$ we have $c \mapsto -c$ and $X \mapsto 1/X$. The one-body part changes as follows: $(N-1)c \mapsto -(N-1)c$; $-\frac{1}{2}(c-1)^2 \mapsto -\frac{1}{2}(c+1)^2$, a change of $-2c$; and $\frac{2NcX}{1-X} \mapsto \frac{-2Nc/X}{1-1/X} = \frac{2Nc}{1-X}$, a change of $+2Nc$. The total change is $-2(N-1)c - 2c + 2Nc = 0$. The term $N^2X/(1-X)^2$ is invariant. In the last fraction, the numerator is multiplied by ξ^{-2N} (its exponents $m, N-m, N+m, 2N-m$ are permuted by $e \mapsto 2N-e$) and so is $(1-X)^2$. The term $m(N-m)/2$ is constant. □

A symbolic check treating ξ, $X = \xi^N$ and $t = \xi^m$ as independent symbols confirms the identity for a_m (`review/palin_symbolic.py`); Lean checks it, and the identities for b_m and g_m (`aBlock_recip`, `bBlock_recip`, `gBlock_recip`).

Remark 6 (geometric meaning). The disk model and its exterior $\{|z| > 1\} \cup \{\infty\}$, each with the metric $4|dz|^2/(1-|z|^2)^2$, are two copies of $\mathbb{H}^2$, exchanged by the inversion $\iota(z) = 1/\bar{z}$ in the boundary circle; ι is an isometry between them. It preserves the pair term, since $\left|\frac{1/\bar{z}-1/\bar{w}}{1-1/(\bar{z}w)}\right| = \frac{|z-w|}{|1-z\bar{w}|}$, and it maps the ring of Euclidean radius $\sqrt{\xi}$ to the ring of radius $1/\sqrt{\xi}$. So the continuation of the blocks to $\xi > 1$ describes the same hyperbolic ring in the exterior chart, and $1/\xi_*$ is the threshold seen from that chart, not a second physical threshold. (We record this interpretation as a remark and do not use it in any proof. The sign change of g_m reflects the fact that ι reverses orientation.)

Clearing denominators, $A_m := 2(1-\xi)^2(1-X)^2 a_m \in \mathbb{Z}[\xi]$ satisfies $A_m(1/\xi) = \xi^{-2N-2}A_m(\xi)$, so A_m is self-reciprocal. We write $B_m := 2(1-X)^2 b_m$, $G_m := 2(1-X)^2 g_m$ and $D_m := A_m B_m - (1-\xi)^2 G_m^2 = 4(1-\xi)^2(1-X)^4(a_m b_m - g_m^2)$. On $(0,1)$ the signs of a_m, b_m and $a_m b_m - g_m^2$ are those of A_m, B_m and D_m.

Threshold equation. If $b_m > 0$ on $(0,1)$ for all m (certified for $N \le 16$, §7; proved for every N, Lemma 12), instability occurs only through the radial entries. The spectral threshold $\xi_*(N)$ is then the largest root in $(0,1)$ of $\prod_m a_m$. For $N \le 16$ this is the unique root of $a_{\lfloor N/2 \rfloor}$ in $(0,1)$. For even N and $m = N/2$, the bracket and the N^2X term combine. With

$$y := \xi^{N/2} = \tanh^N(r/2),$$

the threshold equation $a_{N/2} = 0$ reads

$$(N-1)c - \frac{(c-1)^2}{2} + \frac{2Ncy^2}{1-y^2} = \frac{N^2}{8} + \frac{N^2 y}{2(1-y)^2}, \qquad c = \frac{1+\xi}{1-\xi}. \tag{3}$$

(Use $m\xi^m + (N-m)\xi^{N-m} = Ny$, $(N-m)\xi^{N+m} + m\xi^{2N-m} = Ny^3$, and $y(1+y^2) + 2y^2 = y(1+y)^2$.) For odd N we use the general a_m with $m = (N-1)/2$. In the sinh model the right-hand side is the same Thomson term $N^2/8$ without the image correction, and the left-hand side is $(N-1)(1+c^2)/2$.

Algebraicity is automatic; the structure is not. Since the blocks are rational in ξ with rational coefficients, every threshold is a root of an integer polynomial, so "the thresholds are algebraic" carries no information by itself. What is informative is the structure: the self-reciprocal numerators A_m, the irreducible palindromic factor that carries ξ_*, its degree, and its leading and constant coefficients (§7).

The natural invariant. By reciprocity, if the minimal polynomial f of ξ_* is palindromic of degree n, then $f(x) = x^{n/2} g(x + 1/x)$, and $C := \xi_* + \xi_*^{-1}$ has degree $n/2$ with minimal polynomial g. Since $\xi = \tanh^2(r/2)$,

$$C = \xi_* + \xi_*^{-1} = 2 + \frac{4}{\sinh^2 r_*} = \frac{2(\cosh^2 r_* + 1)}{\cosh^2 r_* - 1}.$$

So $\cosh^2 r_*$ generates the same field as C. In the sinh model, $C = 2 + 4/(q-1)$ is rational (`note.tex`, §4).

7 Certified thresholds for $N \le 16$

Method (`certify_v2.py` → `certify_v2.out`). For each $N \le 16$ and $1 \le m \le \lfloor N/2 \rfloor$:

1. Form the exact integer polynomials A_m, B_m, D_m (sympy, from Thm. 2).
2. Isolate all real roots in $(0,1)$ of the square-free parts with exact rational real-root isolation, after removing the factors ξ and $\xi - 1$. This shows: B_m has no root and $B_m(\frac{1}{2}) > 0$; every A_m with $m \ne \lfloor N/2 \rfloor$ has all its roots strictly below the isolating interval of $A_{\lfloor N/2 \rfloor}$; and $A_{\lfloor N/2 \rfloor}$ has exactly one root in $(0,1)$. Moreover A_1 and A_2 have no root in $(0,1)$ and are positive there, and each A_m with $3 \le m < \lfloor N/2 \rfloor$ has exactly one root in $(0,1)$.
3. Factor A_{crit} over $\mathbb{Q}$ (sympy) and select the unique irreducible factor f with a root in $(0,1)$. Its irreducibility is certified separately with PARI/GP (§7.1). Refine an exact rational isolating interval $[\alpha, \beta]$ for its root, of width $< 10^{-45}$.
4. Assert exactly that $A_{\text{crit}}(\alpha) < 0 < A_{\text{crit}}(\beta)$. With step 2 this proves $a_{\text{crit}} < 0$ on $(0, \xi_*)$ and $a_m > 0$ for all m on $(\xi_*, 1)$.
5. Compute Arb balls for the roots of f (python-flint 0.9, 400 bits). Select the real root by exact tests only: the real roots returned by `arb_fmpz_poly_complex_roots` have imaginary part exactly 0. Assert that the whole ball $[\text{mid} - \text{rad}, \text{mid} + \text{rad}]$, converted exactly to rationals, lies inside $[\alpha, \beta]$.
6. For the energy–momentum threshold: assert that D_1, D_2 have no root in $(0,1)$ and are positive there, and that each D_m with $m \ge 3$ has exactly one root in $(0,1)$ and is positive above it. Take the largest root of $\prod_m D_m$ in $(0,1)$, assert that it is the root of D_{crit}, isolate it exactly, and compare it exactly with $[\alpha, \beta]$. Assert that the root of every other D_m lies below α, and, for odd N, that $D_{\text{crit}}(\beta) < 0$.

No floating-point comparison enters any pass/fail decision. (`certify.py` computes the same values and polynomials but selects the Arb root by a floating-point comparison of its midpoint; no certified statement rests on it.)

Theorem 7 (CERTIFIED, $2 \le N \le 16$)**.** *(a) For $N \le 7$, a_m, b_m and $a_m b_m - g_m^2$ are positive on all of $(0,1)$ for every m. The ring is spectrally stable and has positive-definite non-rotational Hessian at every radius.*

(b) For $8 \le N \le 16$, $b_m > 0$ on $(0,1)$ for every m. The ring is linearly unstable for $0 < \xi < \xi_*(N)$ and spectrally stable for $\xi_*(N) < \xi < 1$. The instability at threshold is in the mode $m = \lfloor N/2 \rfloor$ (and $N - m$). The entries a_1 and a_2 are positive on $(0,1)$, and each a_m with $3 \le m < \lfloor N/2 \rfloor$ has exactly one zero in $(0,1)$, strictly below ξ_*.

(c) (irreducibility certified with PARI/GP, §7.1) $\xi_*(N)$ is the root in $(0,1)$ of an irreducible palindromic primitive integer polynomial f_N of degree

$$\deg \xi_*(N) = \begin{cases} 2N-2, & N \text{ odd}, \\ \frac{3}{2}N-2, & N \equiv 0 \pmod 4, \\ \frac{3}{2}N-3, & N \equiv 2 \pmod 4, \end{cases} \tag{4}$$

with the values in Table 1.

N	ξ_*	$\sqrt{\xi_*}$	r_*	$\cosh r_*$	deg	lead
8	0.06735622300408126226843 74…	0.2595307747	0.5312104560	1.1444414784	10	1
9	0.11287106323660274043969 40…	0.3359628897	0.6990695396	1.2544637167	16	4
10	0.16611424152623911808175 86…	0.4075711490	0.8653901256	1.3984100696	12	7
11	0.20460117688266929063553 58…	0.4523286160	0.9752480466	1.5144618547	20	10
12	0.24741777680406770917396 50…	0.4974110743	1.0917203545	1.6575169309	16	7
13	0.27986138864390668702999 65…	0.5290192706	1.1775646302	1.7772431147	24	18
14	0.31493896311550662339350 54…	0.5611942294	1.2691494599	1.9194478920	18	23
15	0.34245858351217025660102 26…	0.5851996100	1.3406681986	2.0416335000	28	28
16	0.37168181390604545663672 87…	0.6096571282	1.4167509599	2.1831006077	22	17

Table 1: Certified spectral thresholds of the tanh model. $\xi_* = \tanh^2(r_*/2)$ is the squared Euclidean radius in the Poincaré disk, r_* is the hyperbolic radius, "deg" is the degree of the minimal polynomial of ξ_* and "lead" is its leading coefficient (equal to its constant term, by palindromy); both are certified with PARI/GP (§7.1). The Arb radii are $< 3 \cdot 10^{-153}$. The full minimal polynomials, those of $C = \xi_* + 1/\xi_*$, and the discriminants are in `certify_v2.out` and `irreducibility/irreducibility.out`.

Independent confirmation. Full $2N$-dimensional Cartesian spectra, computed without the block formulas (`review/indep_lin.py`, `review/minpoly8.py`), give $\xi_*(8), \xi_*(9), \xi_*(10), \xi_*(12)$ in agreement to ≥ 43 digits. A grid scan of the full Jacobian for $N = 3..7$ gives $\max|\mathrm{Re}\,\sigma| \leq 4 \cdot 10^{-19}$ (`review/small_N.out`; NUMERICAL).

Structure of the numerator. For even $N \in \{8, \ldots, 16\}$, $A_{N/2}$ factors over $\mathbb{Q}$ as a constant times $(\xi-1)^4(\xi^{N/2}+1)$, times one further factor $(\xi+1)$ when $N \equiv 2 \pmod 4$, times the minimal polynomial of ξ_*. So $(\xi+1)^2$ divides $A_{N/2}$ exactly when $N \equiv 2 \pmod 4$, with one copy of $(\xi+1)$ coming from $\xi^{N/2}+1$. (This factorisation is checked in Lean, `A_crit_factor_N8–N16`.) These cyclotomic factors have roots on $|\xi| = 1$ and are spurious.

7.1 Irreducibility (CERTIFIED by PARI/GP, and checked in Lean)

Scripts and output are in `tanh/irreducibility/`. The polynomials f_N ($8 \leq N \leq 16$) are read from `lean/Havelock/Tanh/CertN*.lean`; `extract.py` recomputes them from `exact.py` (factor $A_{\lfloor N/2 \rfloor}$ and take the factor with a root in $(0,1)$) and finds the two sources identical. With PARI/GP 2.15.4 (`irreducibility.gp`) we check, for every N:

1. `polisirreducible`$(f_N) = 1$, content 1, f_N palindromic, and degree and leading coefficient as in Table 1 and (4).

2. A second, independent witness. For the first six primes $p \nmid \operatorname{lead}(f_N)\operatorname{disc}(f_N)$ we record the degrees of the irreducible factors of $f_N \bmod p$. A factorisation over $\mathbb{Q}$ into factors of degrees d and $n-d$ would give a sub-multiset of each pattern summing to d. For every N no $0 < d < n$ is attained by all six patterns, so f_N is irreducible. For example, for $N = 8$ the patterns mod $3, 5, 7$ are $(5,5)$, $(2,8)$, (10).

3. The trace polynomial g_N with $f_N(x) = x^{n/2} g_N(x + 1/x)$ is irreducible and agrees with `certify_v2.out`; the discriminants agree too.

Since f_N is irreducible and primitive, it is (up to sign) the minimal polynomial of $\xi_*(N)$ over $\mathbb{Z}$. If $\operatorname{lead}(f_N) \neq \pm 1$, then by Gauss's lemma the monic minimal polynomial $f_N/\operatorname{lead}(f_N)$ is not in $\mathbb{Z}[x]$, so $\xi_*(N)$ is not an algebraic integer.

Mod-p certificates. If f_N is irreducible mod a prime $p \nmid \operatorname{lead}(f_N)$, then f_N is irreducible over $\mathbb{Q}$. Such a prime exists for every N here. The smallest ones are $p = 7, 13, 107, 17, 11, 101, 3, 29, 163$ for $N = 8, \ldots, 16$. These certificates are checked in Lean (`lean/Havelock/TanhIrred.lean`, theorems `f8_irreducible`, …, `f16_irreducible`: irreducibility over $\mathbb{Z}$ and $\mathbb{Q}$, by kernel reduction). Lean proves irreducibility only; the consequences above (minimal polynomial, degree, and, via Gauss's lemma, that $\xi_*(N)$ is not an algebraic integer for $N \geq 9$) are on paper.

7.2 $N = 8$: an algebraic unit

Corollary 8 (CERTIFIED; irreducibility by PARI/GP)**.** *$\xi_*(8)$ is an algebraic unit of degree 10 with minimal polynomial*

$$x^{10} - 12x^9 - 37x^8 - 72x^7 - 84x^6 - 88x^5 - 84x^4 - 72x^3 - 37x^2 - 12x + 1.$$

$C = \xi_* + \xi_*^{-1} = 2 + 4/\sinh^2 r_*$ has minimal polynomial $C^5 - 12C^4 - 42C^3 - 24C^2 + 32C + 32$. For $9 \leq N \leq 16$ the leading coefficients of the (irreducible, primitive) f_N are $4, 7, 10, 7, 18, 23, 28, 17$, so $\xi_*(N)$ is not an algebraic integer.

The polynomial is irreducible (§7.1), monic, with constant term 1, so $\xi_*(8)$ is a unit. PSLQ applied to a 110-digit value from the independent Cartesian computation recovers the same polynomial (`review/minpoly8.py`). Numerically, its only real roots are $\xi_*(8)$ and $1/\xi_*(8) \approx 14.846$. Four conjugates lie on $|x| = 1$ and four do not, so $\xi_*(8)$ is neither Pisot nor Salem. In the sinh model $\xi_*(8) = 8 - 3\sqrt{7}$ is also a unit, of degree 2. We do not know whether the coincidence has a reason.

Conjecture 9. *For every $N \geq 8$: the factor of $A_{\lfloor N/2 \rfloor}$ carrying ξ_* is irreducible and palindromic, so that $1/\xi_*$ is a Galois conjugate of ξ_*; and its degree is given by (4).*

Reciprocity alone shows that $1/\xi_*$ is a root of $A_{\lfloor N/2 \rfloor}$. It does not show that $1/\xi_*$ is a conjugate of ξ_*. That needs irreducibility of the factor, which is why palindromy of the minimal polynomial is certified only for $N \leq 16$. ($b_m > 0$ and the critical mode $\lfloor N/2 \rfloor$ are proved for every N, Lemmas 12 and 13.)

8 Odd N: the Krein window

For even N, $g_{N/2} = 0$, so $D_{N/2} = A_{N/2}B_{N/2}$ and the two thresholds coincide. This is certified: the same irreducible factor carries both. For odd N, $g_{\text{crit}} \neq 0$. At ξ_* we have $a_{\text{crit}} b_{\text{crit}} - g_{\text{crit}}^2 = -g_{\text{crit}}^2 < 0$, and the Hessian becomes definite only at a larger value ξ_{EM}.

Proposition 10 (CERTIFIED, odd $9 \le N \le 15$)**.** $\xi_{\mathrm{EM}}(N) > \xi_*(N)$ *strictly (disjoint exact rational intervals). The gaps are* $3.220 \cdot 10^{-8}$ $(N = 9)$*,* $1.244 \cdot 10^{-7}$ (11)*,* $1.777 \cdot 10^{-7}$ (13) *and* $1.886 \cdot 10^{-7}$ (15)*. On* $(\xi_*, \xi_{\mathrm{EM}})$ *the ring is linearly (spectrally) stable, with distinct purely imaginary eigenvalues, but the critical block* M_{crit} *has one positive and one negative eigenvalue, while every other block is positive definite. Its nonlinear stability there is undecided.*

In the window the two frequencies of the critical block carry opposite Krein signatures. A check on the coordinate-free Cartesian Hessian (count of negative eigenvalues, Krein signs of the eigenvectors; `review/krein_em.py`) confirms this, so it is not a coordinate artefact. The energy–momentum method gives no conclusion there: the ring is linearly (spectrally) stable, and nonlinear stability is **undecided**. In the sinh model $g_m \equiv 0$, and no such window exists.

9 Large N: the continuum limit and a critical spacing

This section proves that $\cosh r_*/N$ converges, identifies the limit, and interprets it. The key is an exact identity: in a suitable radial variable the blocks of Thm. 2 are, up to a term that does not depend on m, a function of m/N alone. Throughout, put

$$\tau := -\tfrac{N}{2}\ln\xi = N\,\mathrm{artanh}\,\tfrac{1}{c} \in (0,\infty), \qquad x := 1 - \tfrac{2m}{N} \in (-1,1), \qquad \theta := \tfrac{m}{N} = \tfrac{1-x}{2},$$

so that $\xi^N = e^{-2\tau}$, $c = \cosh r = \coth(\tau/N)$, and $\tau \to 0$ as $r \to \infty$. Let $Q(x,\tau) := \cosh\tau\cosh(x\tau) - x\sinh\tau\sinh(x\tau)$, and let $\mathcal{L}(\eta) := \coth\eta - 1/\eta$ be the Langevin function.

Theorem 11 (exact symbol; PROVED)**.** *For every* $N \ge 2$*,* $1 \le m \le N-1$ *and* $\xi \in (0,1)$*, with* $k := c/N = \coth(\tau/N)/N$*,*

$$\frac{b_m}{N^2} = \beta(x,\tau) := \frac{1-x^2}{8} + \frac{1-Q(x,\tau)}{4\sinh^2\tau},$$
$$\frac{a_m}{N^2} = E_N(\tau) - \frac{1}{4\sinh^2\tau} - \psi(x,\tau), \qquad \psi(x,\tau) := \frac{1-x^2}{8} + \frac{Q(x,\tau)}{4\sinh^2\tau}, \qquad E_N(\tau) := k\coth\tau - \frac{k^2}{2} - \frac{1}{2N^2},$$
$$\frac{g_m}{N^2} = \gamma(x,\tau) := \frac{1}{2}\Big[\frac{\theta e^{x\tau} - (1-\theta)e^{-x\tau}}{2\sinh\tau} + \frac{e^{-\tau}\sinh(x\tau)}{2\sinh^2\tau}\Big].$$

The dependence on m *enters only through* x*, and in* a_m *only through* ψ*.*

Proof. Substitute $\xi^j = e^{-2j\tau/N}$, so $\xi^m = e^{-(1-x)\tau}$, $\xi^{N-m} = e^{-(1+x)\tau}$, $1 - X = 2e^{-\tau}\sinh\tau$. From the closed form of W_k, $W_m/N = \theta e^{x\tau}/(2\sinh\tau) + e^{-\tau}e^{x\tau}/(4\sinh^2\tau)$ and $W_{N-m}/N = (1-\theta)e^{-x\tau}/(2\sinh\tau) + e^{-\tau}e^{-x\tau}/(4\sinh^2\tau)$, and $NW_N = N^2/(4\sinh^2\tau)$. Since $\theta e^{x\tau} + (1-\theta)e^{-x\tau} = \cosh(x\tau) - x\sinh(x\tau)$ and $\sinh\tau + e^{-\tau} = \cosh\tau$,

$$\frac{W_m + W_{N-m}}{N} = \frac{\sinh\tau\,(\cosh x\tau - x\sinh x\tau) + e^{-\tau}\cosh x\tau}{2\sinh^2\tau} = \frac{Q(x,\tau)}{2\sinh^2\tau}.$$

With $m(N-m) = N^2(1-x^2)/4$ this gives b_m and g_m. For a_m use the form $a_m = V''(0) - \frac{m(N-m)}{2} - NW_N - \frac{N}{2}(W_m + W_{N-m})$ from the proof of Thm. 2, where $V''(0) = (N-1)c - \frac{1}{2}(c-1)^2 + \frac{2NcX}{1-X} = Nc\,\frac{1+X}{1-X} - \frac{c^2}{2} - \frac{1}{2}$ and $\frac{1+X}{1-X} = \coth\tau$. □

The identity holds exactly; `large_n.py` confirms it against `blocks.py` at 400 random (N, m, ξ), $N \le 300$, to $8 \cdot 10^{-59}$.

Lemma 12 (tangential stiffness; PROVED)**.** *Put $a=(1+x)\tau$, $b=(1-x)\tau$ and $f(t)=\sinh^2(t/2)/t$. Then*

$$\beta(x,\tau)=\frac{ab}{4\tau\sinh^2\tau}\big[f(a+b)-f(a)-f(b)\big]>0\qquad(|x|<1,\ \tau>0).$$

Hence $b_m>0$ for every $N\ge 2$, every $1\le m\le N-1$ and every radius.

Proof. $1-x=b/\tau$, $1+x=a/\tau$ and $a+b=2\tau$ give $2Q=(b\cosh a+a\cosh b)/\tau=2+2\big(b\sinh^2\frac{a}{2}+a\sinh^2\frac{b}{2}\big)/\tau$ and $(1-x^2)\sinh^2\tau=ab\sinh^2\tau/\tau^2=2ab\,f(a+b)/\tau$. Substituting into $8\sinh^2\tau\,\beta=(1-x^2)\sinh^2\tau+2-2Q$ gives the formula. The function $f(t)/t=(\sinh(t/2)/t)^2$ is strictly increasing, so $f(a+b)=a\frac{f(a+b)}{a+b}+b\frac{f(a+b)}{a+b}>a\frac{f(a)}{a}+b\frac{f(b)}{b}=f(a)+f(b)$ for $a,b>0$. □

Lemma 13 (critical mode; PROVED)**.** *For every $\tau>0$, $\psi(\cdot,\tau)$ is even and strictly decreasing on $[0,1]$. Hence, for every $N\ge 2$ and every radius, $a_m\ge a_{\lfloor N/2\rfloor}$, with equality only for $m\in\{\lfloor N/2\rfloor, N-\lfloor N/2\rfloor\}$.*

Proof. Evenness is clear. Write $h=x\tau\in(0,\tau]$, $T=\tau\coth\tau$ and $S(t)=\sinh t/t$. A direct computation gives

$$4\sinh^2\tau\,\partial_x\psi=\sinh\tau\,v(h),\qquad v(h):=(T-1)\sinh h-h\cosh h-h\,S(\tau),$$

and $v(h)/\sinh h=(T-1)-h\coth h-S(\tau)/S(h)$. So we must show $T-1-h\coth h<S(\tau)/S(h)$ for $0<h\le\tau$. Let $\lambda(t)=\ln S(t)$, so $\lambda'=\coth t-1/t$, $\lambda''=t^{-2}-\sinh^{-2}t\ge 0$, and $(t\coth t)'=\lambda'+t\lambda''$. With $D:=\lambda(\tau)-\lambda(h)\ge 0$ and $I:=\int_h^\tau t\lambda''(t)\,dt\ge 0$ we have $T-h\coth h=D+I$, and the claim becomes

$$I<1+e^D-D.$$

The right side is increasing in $D\ge 0$ and exceeds 2 when $h<\tau$ (then $D>0$); at $h=\tau$, $I=0$. We use $\lambda''(t)\le\min(\frac13,t^{-2})$. (The bound $\frac13$ is $3(\sinh^2 t-t^2)\le t^2\sinh^2 t$; compare the Taylor coefficients of t^{2j}, which are $3\cdot 2^{2j-1}/(2j)!$ and $2^{2j-3}/(2j-2)!$, with ratio $(2j)(2j-1)/12\ge 1$ for $j\ge 2$.) Hence $I\le\int_0^\tau t\min(\frac13,t^{-2})\,dt$, which is $\tau^2/6$ for $\tau\le\sqrt{3}$ and $\frac12+\ln(\tau/\sqrt{3})$ otherwise; both are ≤ 2 when $\tau\le\sqrt{3}\,e^{3/2}\approx 7.76$. For $\tau>\sqrt{3}\,e^{3/2}$: if $h\ge\tau/2$ then $I\le\int_{\tau/2}^\tau dt/t=\ln 2<2$; if $h<\tau/2$ then $e^D\ge S(\tau)/S(\tau/2)=\cosh(\tau/2)$, so the right side is at least $1+\frac12 e^{\tau/2}-\frac{\tau}{2}$, which exceeds $\frac12+\ln(\tau/\sqrt{3})\ge I$ for $\tau\ge 7.76$. So $\partial_x\psi<0$ on $(0,1]$. The statement about a_m follows from Thm. 11, since $|x|=|1-2m/N|$ is smallest at $m=\lfloor N/2\rfloor$ and $N-\lfloor N/2\rfloor$. □

Corollary 14 (PROVED, all N)**.** *For every $N\ge 2$ and every radius, the ring is spectrally stable if and only if $a_{\lfloor N/2\rfloor}\ge 0$; when $a_{\lfloor N/2\rfloor}<0$ the modes $m=\lfloor N/2\rfloor$ and $N-\lfloor N/2\rfloor$ grow exponentially. This removes the restriction $N\le 16$ from the statements "$b_m>0$" and "the critical mode is $\lfloor N/2\rfloor$" of Thm. 7(b).*

Proof. Prop. 3 with Lemmas 12 and 13. □

Proposition 15 (even N: one threshold; PROVED)**.** *Let N be even, $\eta=\tau/2$ and $\mu=\frac{2}{N}\coth\frac{2\eta}{N}$. Then*

$$\frac{8a_{N/2}}{N^2}=2\mu\tanh\eta-(\coth\eta-\mu)^2-\frac{4}{N^2},$$

which for $N\ge 4$ is strictly decreasing in η, from $2-4/N^2$ ($r\to\infty$) to $-(N^2-8N+8)/N^2$ ($r\to 0$). Hence every even $N\le 6$ is stable at every radius, and every even $N\ge 8$ has exactly one spectral threshold, at a simple sign change of $a_{N/2}$.

Proof. At $x = 0$, $\psi + \frac{1}{4\sinh^2\tau} = \frac{1}{8} + \frac{1}{8\sinh^2\eta}$; with $\coth\tau = \frac{1}{2}(\coth\eta + \tanh\eta)$, $k = \mu/2$ and $\sinh^{-2}\eta = \coth^2\eta - 1$ the formula follows from Thm. 11. For $N = 2$, $\mu = \coth\eta$ and the expression is identically 1, so the ring is stable at every radius. For $N \geq 4$, with $v = 2\eta/N$ and $q = N/2 > 1$, $\mu\tanh\eta = \frac{2}{N}\tanh(qv)/\tanh v$, and $\frac{d}{dv}\ln\frac{\tanh(qv)}{\tanh v} = \frac{2q}{\sinh 2qv} - \frac{2}{\sinh 2v} < 0$, because $t/\sinh t$ is strictly decreasing. Since $t\coth t$ is increasing, $\mu \leq \coth\eta$; and $\frac{d}{d\eta}(\coth\eta - \mu) = \frac{(2/N)^2}{\sinh^2(2\eta/N)} - \frac{1}{\sinh^2\eta} \geq 0$, again because $t/\sinh t$ decreases. So $(\coth\eta - \mu)^2$ is non-decreasing and the whole expression is strictly decreasing. The limits are direct. □

At $r \to 0$ the limit $-(N^2 - 8N + 8)/N^2$ is the planar (Thomson–Havelock) value, negative exactly for $N \geq 7$. For odd N the same computation gives $8a_{(N-1)/2}/N^2 = P_{12} + P_3 - 3/N^2$, where $P_{12} = 2\mu\tanh\eta - (\coth\eta - \mu)^2$ and $P_3 = \left[\frac{2}{N}\sinh v\sinh Nv - 2\cosh Nv(\cosh v - 1)\right]/\sinh^2 Nv$, $v = \tau/N$; its planar limit is $-(N-1)(N-7)/N^2$. The term P_3 is not monotone, but where it increases its slope is below 0.61% of $|P_{12}'|$ for the sampled odd $N \in \{9, 11, 15, 21, 51, 101, 1001\}$, and below 37.5% for $N = 3$ (`large_n.out`, NUMERICAL). For odd $N \geq 17$ the threshold is nevertheless unique: see Prop. 21 at the end of this section.

The limit. Let $E_\infty(\tau) = \coth\tau/\tau - 1/(2\tau^2)$ and $\Phi(\tau) := E_\infty(\tau) - \frac{1}{4\sinh^2\tau} - \psi(0,\tau)$, the $N \to \infty$ limit of $a_{\lfloor N/2\rfloor}/N^2$. With $\eta = \tau/2$,

$$8\,\Phi = \frac{2\tanh\eta}{\eta} - \mathcal{L}(\eta)^2. \tag{5}$$

Since $(\tanh\eta/\eta)' = (\eta - \sinh\eta\cosh\eta)/(\eta^2\cosh^2\eta) < 0$ and $(\mathcal{L}^2)' = 2\mathcal{L}\mathcal{L}' > 0$ ($\mathcal{L} > 0$, $\mathcal{L}' = \eta^{-2} - \sinh^{-2}\eta > 0$), we have $\Phi' < 0$ everywhere: Φ decreases strictly from $\frac{1}{4}$ at $\tau = 0$ to $-\frac{1}{8}$ at $\tau = \infty$. Let η_∞ be its unique zero:

$$\Big(\coth\eta_\infty - \frac{1}{\eta_\infty}\Big)^2 = \frac{2\tanh\eta_\infty}{\eta_\infty}, \qquad \eta_\infty = 3.7226696224745093436\ldots, \qquad \kappa_\infty := \frac{1}{2\eta_\infty} = 0.13431221427262813905 \tag{6}$$

Dropping the exponentially small terms ($\tanh\eta, \coth\eta \to 1$) turns (6) into $\eta^2 - 4\eta + 1 = 0$, so $\eta_\infty \approx 2 + \sqrt{3}$ (off by $9.4 \cdot 10^{-3}$) and $\kappa_\infty \approx 1 - \sqrt{3}/2$ (off by $3.4 \cdot 10^{-4}$).

Theorem 16 (large-N limit; PROVED)**.** *Let $r_*(N)$ be the spectral threshold radius, i.e. $\xi_*(N) = \inf\{\xi_0 :$ the ring is spectrally stable for all $\xi \in (\xi_0, 1)\}$. Then, for both parities,*

$$\cosh r_*(N) = \kappa_\infty N + O(1/N), \qquad \ell_N := \frac{2\pi\sinh r_*(N)}{N} = \ell_* + O(N^{-2}), \qquad \ell_* := 2\pi\kappa_\infty = \frac{\pi}{\eta_\infty} = 0.8439085\ldots$$

In words: at threshold, the arc length along the ring between neighbouring vortices tends to $\ell_* \approx 0.844$ curvature radii. (The geodesic distance between neighbours tends to $2\,\mathrm{arsinh}(\ell_*/2) \approx 0.821$.)

Proof. Let $\tau_*(N) = -\frac{N}{2}\ln\xi_*(N)$. By Thm. 11, $a_{\lfloor N/2\rfloor}/N^2 = \Phi(\tau) + \Delta_N(\tau) + \varepsilon_N(\tau)$ with $\Delta_N = E_N - E_\infty$ and $\varepsilon_N = \psi(0,\tau) - \psi(x_N,\tau)$, where $x_N = 0$ (N even) or $1/N$ (N odd). By Lemma 13, $\varepsilon_N \geq 0$; by Taylor's theorem $\varepsilon_N(\tau) \leq \frac{1}{2N^2}\max_{|x|\leq 1/N}|\psi_{xx}(x,\tau)|$, bounded by C/N^2 for τ in a compact set. For Δ_N, put $\delta = k - 1/\tau = \tau^{-1}(u\coth u - 1)$, $u = \tau/N$; then $0 < \delta < \tau/(3N^2)$ and $\Delta_N = \delta(\coth\tau - \frac{1}{\tau} - \frac{\delta}{2}) - \frac{1}{2N^2}$. Since $0 < \coth\tau - 1/\tau < 1$, $-\frac{1}{2N^2} - \frac{\delta^2}{2} \leq \Delta_N \leq \frac{\tau}{3N^2}$, so $|\Delta_N| \leq C_T/N^2$ on $(0,T]$, uniformly down to $\tau \to 0$. Fix $T > \tau_\infty = 2\eta_\infty$ and let $c_0 = \frac{1}{2}|\Phi'(\tau_\infty)| > 0$. For $0 < d \leq d_0$ we have $\Phi(\tau_\infty - d) \geq c_0 d$ and $\Phi(\tau_\infty + d) \leq -c_0 d$. Take $d = K/N^2$ with K large. On $(0, \tau_\infty - d]$, $a_{\lfloor N/2\rfloor}/N^2 \geq c_0 d - C_T/N^2 > 0$, so every $a_m > 0$ (Lemma 13) and every $b_m > 0$

(Lemma 12): the ring is spectrally stable. At $\tau_\infty + d$, $a_{\lfloor N/2\rfloor}/N^2 \le -c_0 d + (C_T + C)/N^2 < 0$ while $b_{\lfloor N/2\rfloor} > 0$, so by Prop. 3 the ring is unstable. Hence $|\tau_*(N) - \tau_\infty| \le K/N^2$ for large N. Finally $\cosh r_* = \coth(\tau_*/N) = N/\tau_* + O(1/N)$ and $\sinh r_* = 1/\sinh(\tau_*/N)$, so $\ell_N = 2\pi/(N\sinh(\tau_*/N)) = 2\pi/\tau_* + O(N^{-2})$. □

Also $r_* = \ln(2\kappa_\infty N) + o(1)$ and $1 - \xi_* \sim 2/(\kappa_\infty N) \approx 14.89/N$. The second-order expansion below gives $\ell_N - \ell_* \approx 2\pi(C_1 - \frac{1}{2\kappa_\infty})/N^2 \approx -20/N^2$: finite rings reach stability at a slightly smaller spacing.

Second order (DERIVED, NUMERICAL check). Expanding $\Delta_N = \frac{1}{N^2}\big[\frac{\tau\coth\tau - 1}{3} - \frac{1}{2}\big] + O(N^{-4})$ and $\varepsilon_N = -\psi_{xx}(0,\tau)/(2N^2) + O(N^{-4})$ (odd N) in the threshold equation gives

$$\cosh r_* = \kappa_\infty N + \frac{C_1}{N} + O(N^{-3}), \qquad C_1 = \frac{\tau_\infty}{3} - \frac{\mathcal{D}}{|\Phi'(\tau_\infty)|\,\tau_\infty^2},$$

with $\mathcal{D} = \frac{\tau_\infty\coth\tau_\infty - 1}{3} - \frac{1}{2}$ for even N and $\mathcal{D} - \frac{1}{2}\psi_{xx}(0,\tau_\infty)$ for odd N. Numerically $C_1^{\text{even}} = 0.5412188400$ and $C_1^{\text{odd}} = 0.4010382403$. The two values differ. Against the certified and the computed thresholds, $(\cosh r_* - \kappa_\infty N)N$ equals 0.54122060 at $N = 800$ and 0.40103931 at $N = 801$ (`large_n.out`). We have not written out the remainder bounds for this second-order statement.

The energy–momentum threshold. Positive definiteness of the Hessian on the non-rotational modes requires $a_m b_m - g_m^2 > 0$ for every m (Prop. 3). Let $\Phi_N(\tau) := E_N(\tau) - \frac{1}{4\sinh^2\tau} - \psi(0,\tau)$, so that $a_m/N^2 = \Phi_N + \psi(0,\tau) - \psi(x,\tau)$ and, for even N, $\Phi_N = a_{N/2}/N^2$. Put $R(x,\tau) := \cosh\tau\sinh(x\tau) - x\sinh\tau\cosh(x\tau)$; then $\gamma = R/(4\sinh^2\tau)$ (as in the proof of Thm. 11, using $\theta e^{x\tau} - (1-\theta)e^{-x\tau} = \sinh x\tau - x\cosh x\tau$).

Lemma 17 (PROVED)**.** *Let $G(x,\tau) := \psi(0,\tau) - \psi(x,\tau) - \gamma(x,\tau)^2/\beta(x,\tau)$ for $|x| < 1$. Then*

$$\beta\, G = \frac{(\cosh\tau + 1)^2}{32\sinh^4\tau}\, K(x,\tau), \qquad K := \cosh\tau\,(1 - \cosh x\tau) + x\sinh\tau\sinh x\tau - \frac{x^2(\cosh\tau - 1)}{2}\big[2 - (1-x^2)(\cosh\tau - 1)\big],$$

and $K > 0$ for $0 < |x| < 1$, $\tau > 0$. Hence $G \ge 0$, with equality only at $x = 0$, and for every N, m and radius

$$a_m b_m - g_m^2 = N^4\,\beta(x,\tau)\,\big[\Phi_N(\tau) + G(x,\tau)\big].$$

Proof. Write $C = \cosh\tau$, $S = \sinh^2\tau$. Then $8S(\psi(0) - \psi(x)) = A := x^2 S + 2C - 2Q$, $8S\beta = B := (1-x^2)S + 2 - 2Q > 0$ (Lemma 12) and $\gamma^2/\beta = R^2/(2SB)$, so $8SB\,G = AB - 4R^2$. Expanding gives $Q^2 - R^2 = C^2 - x^2 S$, so $AB - 4R^2$ is linear in Q; using $S + 2C + 2 = (C+1)^2$ one finds $AB - 4R^2 = 2(C+1)^2 K$. For the sign, $K(x,0) = 0$ and

$$\partial_\tau K = \sinh\tau\,(1-x^2)\big[x^2(\cosh\tau - 1) - (\cosh x\tau - 1)\big] > 0 \qquad (0 < |x| < 1,\ \tau > 0),$$

because $\cosh x\tau - 1 = \sum_{k\ge1} x^{2k}\tau^{2k}/(2k)! < x^2\sum_{k\ge1}\tau^{2k}/(2k)!$. With $\beta > 0$ this gives $G \ge 0$. The last identity combines $a_m/N^2 = \Phi_N + \psi(0) - \psi(x)$ with $b_m/N^2 = \beta$ and $g_m/N^2 = \gamma$. □

Theorem 18 (energy–momentum threshold; PROVED)**.** *(a) For every even N and every radius, the Hessian is positive definite on the non-rotational modes if and only if $a_{N/2} > 0$. Hence for every even N the energy–momentum threshold equals the spectral threshold, and there is no Krein window. (For $N = 2, 4, 6$ there is no threshold at all: the ring is stable, with definite Hessian, at every radius.)*

(b) For every odd $N \geq 9$ the Krein window is non-empty: $\xi_{\mathrm{EM}}(N) > \xi_*(N)$; the ring is spectrally stable on $(\xi_*, 1)$, and the critical block is indefinite on a right-neighbourhood of ξ_*. Moreover $\xi_{\mathrm{EM}}(N) - \xi_*(N) = O(N^{-3})$.

(c) For both parities, $\cosh r_{\mathrm{EM}}(N) = \kappa_\infty N + O(1/N)$.

Here $\xi_{\mathrm{EM}}(N) = \inf\{\xi_0 :$ *the Hessian is positive definite on the non-rotational modes for all* $\xi \in (\xi_0, 1)\}$.

Proof. (a) $a_m \geq a_{N/2} = N^2\Phi_N$ (Lemma 13). If $a_{N/2} > 0$, Lemma 17 gives $a_m b_m - g_m^2 > 0$ for every m; if $a_{N/2} \leq 0$ the Hessian is not positive definite. Spectral stability is $a_{N/2} \geq 0$ (Cor. 14). By Prop. 15, $a_{N/2}$ has no zero for $N \leq 6$ and a single simple sign change for $N \geq 8$, so the two thresholds coincide. (b) For odd $N \geq 9$ the planar limit $-(N-1)(N-7)/N^2 < 0$ of $8a_{\mathrm{crit}}/N^2$ and its limit $2 - 2/N^2 > 0$ as $r \to \infty$ show that $\xi_* \in (0,1)$ exists, and $a_{\mathrm{crit}}(\xi_*) = 0$. There $a_{\mathrm{crit}} b_{\mathrm{crit}} - g_{\mathrm{crit}}^2 = -g_{\mathrm{crit}}^2 < 0$, because $R(1/N, \tau) > 0$: $R/(\cosh\tau \cosh x\tau) = \tanh x\tau - x\tanh\tau > 0$ by strict concavity of tanh on $[0, \infty)$. By continuity the Hessian is indefinite on a neighbourhood of ξ_*, so $\xi_{\mathrm{EM}} > \xi_*$; by the definition of ξ_* the ring is spectrally stable on $(\xi_*, 1)$. For the width, Lemma 17 shows the Hessian is positive definite wherever $\Phi_N > 0$ (then also every $a_m > 0$). In the proof of Thm. 16, $\Phi_N = \Phi + \Delta_N$ and $\Phi_N + \varepsilon_N = a_{\mathrm{crit}}/N^2$, with $|\Delta_N|, \varepsilon_N = O(N^{-2})$ near τ_∞. Also $|\Delta_N'| = O(N^{-2})$: $\delta = \mathcal{L}(\tau/N)/N$, so $0 < \delta' = \mathcal{L}'(\tau/N)/N^2 \leq 1/(3N^2)$, and differentiating $\Delta_N = \delta(\coth\tau - \frac{1}{\tau} - \frac{\delta}{2}) - \frac{1}{2N^2}$ gives $|\Delta_N'| \leq (1 + \tau/3)/(3N^2)$. Since $|\Phi'| \geq \frac{3}{2}c_0$ near τ_∞ (shrinking d_0 if needed), $\Phi_N' \leq -c_0$ there for large N. So Φ_N is positive below its first zero τ_Φ and strictly decreasing near it, and $\tau_\Phi \leq \tau_{\mathrm{EM}} \leq \tau_*$ with $\tau_* - \tau_\Phi \leq \varepsilon_N(\tau_*)/c_0 = O(N^{-2})$. Since $\xi = e^{-2\tau/N}$, $\xi_{\mathrm{EM}} - \xi_* \leq \frac{2}{N}(\tau_* - \tau_\Phi) = O(N^{-3})$. (c) τ_Φ and τ_* are both $\tau_\infty + O(N^{-2})$, so τ_{EM} is too, and the conversion is as in Thm. 16. □

Part (a) extends to all even N what Thm. 7 certifies for $N \leq 16$. Part (b) extends the non-emptiness in Prop. 10 (certified for $N = 9, 11, 13, 15$) to every odd $N \geq 9$. That the critical block stays indefinite on the whole window, with every other block definite, is certified for $N \leq 15$ (Prop. 10) and checked numerically for $N \in \{17, 21, 31, 61, 121, 241\}$; it is not proved in general. The bound in (b) is crude: the intermediate bound $\xi_{\mathrm{EM}} \leq \xi_\Phi := e^{-2\tau_\Phi/N}$ gives $N^3(\xi_{\mathrm{EM}} - \xi_*) \leq 14.62$ at $N = 241$, while the actual value is 0.00348. Expanding the critical block alone suggests $N^3(\xi_{\mathrm{EM}} - \xi_*) \to W = 2\gamma_x(0,\tau_\infty)^2/(\beta(0,\tau_\infty)|\Phi'(\tau_\infty)|) \approx 0.0037$; the data (0.0028 at $N = 61$, 0.0032 at $N = 121$) approach it slowly. That constant is DERIVED, not proved: it assumes that the critical mode is the last to become definite.

Physical interpretation

A critical spacing. Theorem 16 says that a large tanh-model ring becomes spectrally stable once neighbouring vortices are more than $\ell_* = 0.8439\,R$ apart in arc length along the ring (asymptotically), where R is the curvature radius (here $R = 1$). Equivalently, the line density $N/(2\pi\sinh r)$ must be below $1/\ell_* \approx 1.185/R$. The approximation $\eta_\infty \approx 2 + \sqrt{3}$ gives $\ell_* \approx \pi(2 - \sqrt{3})R = 0.8418R$. In the sinh model there is no such scale: $\cosh r_* \sim \sqrt{N}/2$, so the threshold spacing $\approx \pi/\sqrt{N}$ tends to 0 (§10).

The limit object is a vortex row on a horocycle. As $N \to \infty$ at fixed τ, the ring radius tends to infinity while the spacing $2\pi\sinh r/N \to 2\pi/\tau$ stays fixed, and the geodesic curvature $\coth r$ of the ring tends to 1, that of a horocycle. In the upper half-plane (curvature -1, area form $dx\,dy/y^2$) the tanh pair term is $-\ln\tanh(d/2) = -\ln|z - w| + \ln|z - \bar{w}|$. Take the horocycle $\mathrm{Im}\, z = 1$ and the infinite row $z_k = k\ell + i$, $k \in \mathbb{Z}$. The x-translations are isometries with momentum

$J = -\sum_j 1/y_j$; put $F = H - vJ$. (The pair term and its image each pull like $1/k$, but together their force decays like k^{-3} in x and k^{-2} in y, so the equations of motion of the row converge absolutely; H and J themselves are infinite and enter only through these.) The Hessian of F at the row couples sites through absolutely summable terms (each of order $1/k^2$), so on the Bloch wave $(\delta x_k, \delta y_k) = e^{2\pi i\theta k}(X, Y)$ it acts as a Hermitian 2×2 matrix $M(\theta)$, $0 \le \theta \le 1$.

Proposition 19 (horocyclic row; PROVED)**.** *The row is a relative equilibrium, translating with $v = \frac{1}{2}(\tau\coth\tau - 1)$, where $\tau := 2\pi/\ell$. Its Bloch symbol is the large-N limit of the ring blocks:*

$$\Big(\frac{\ell}{2\pi}\Big)^2 M(\theta) = \begin{pmatrix} \beta & i\gamma \\ -i\gamma & \alpha \end{pmatrix}(x,\tau), \qquad x = 1-2\theta, \qquad \alpha := E_\infty - \frac{1}{4\sinh^2\tau} - \psi,$$

in the basis $(\delta x, \delta y)$. So, after rescaling (N^{-2} for the ring, $(\ell/2\pi)^2$ for the row), the ring's mode number m/N becomes the Bloch wavenumber θ; the ring's tangential and outward radial directions correspond to δx and $-\delta y$. This correspondence is orientation-preserving: it keeps θ and the sign of γ.

Proof. Write $p_k = 1/(k\ell - 2i)^2$, so that $p_{-k} = \bar p_k$, and $b = 2/\ell = \tau/\pi$. For $h = \operatorname{Re} f$ with f holomorphic, $\operatorname{Hess} h = \left(\begin{smallmatrix} \operatorname{Re} f'' & -\operatorname{Im} f'' \\ -\operatorname{Im} f'' & -\operatorname{Re} f'' \end{smallmatrix}\right)$. Applied to $-\ln|z_0 - z_k|$ and $\ln|z_0 - \bar z_k|$ (in the second, $\partial_{x_k} = -\partial_x$ and $\partial_{y_k} = +\partial_y$), the on-site block is $\operatorname{diag}(S_1, -S_1) + \sum_{k\ne0} \operatorname{Re} p_k \operatorname{diag}(-1,1) + \operatorname{diag}(0, 2v)$, with $S_1 = \sum_{k\ne0}(k\ell)^{-2}$ (the off-diagonal terms cancel in pairs), and the coupling block to site k is $\operatorname{diag}(-1,1)(k\ell)^{-2} + \left(\begin{smallmatrix} \operatorname{Re} p_k & \operatorname{Im} p_k \\ -\operatorname{Im} p_k & \operatorname{Re} p_k \end{smallmatrix}\right)$. Force balance in y gives $v = \sum_{k\ne0} 2/(k^2\ell^2 + 4)$. Four classical sums evaluate everything, for $0 \le q := 2\pi\theta \le 2\pi$:

$$\sum_{k\ne0} k^{-2} = \pi^2/3, \qquad \sum_{k\ge1}\cos(qk)/k^2 = \pi^2/6 - \pi q/2 + q^2/4, \qquad \sum_{k\in\mathbb{Z}}(k^2+a^2)^{-1} = (\pi/a)\coth\pi a,$$
$$\sum_{k\in\mathbb{Z}} e^{iqk}(k-ib)^{-2} = -2\pi\sum_{n\ge0}(q+2\pi n)\,e^{-b(q+2\pi n)} \quad (b>0).$$

The last is Poisson summation, $\sum_k f(k) = \sum_n \hat f(2\pi n)$ with $\hat f(\omega) = \int f(t)e^{-i\omega t}dt$, which applies because $f(t) = e^{iqt}(t-ib)^{-2} = O(t^{-2})$ and $\hat f$ is continuous and decays exponentially: $\hat f(2\pi n)$ is $-2\pi(q-2\pi n)e^{-b(q-2\pi n)}$ for $q \ge 2\pi n$ (residue at $t = ib$) and 0 otherwise; the terms with $q = 2\pi n$ vanish, so the formula holds on the closed interval $0 \le q \le 2\pi$. With $\pi b = \tau$ and $X = e^{-2\tau}$, the third sum gives v, and the fourth gives $P(\theta) := \sum_{k\ne0} e^{iqk}p_k = \frac14 - \tau^2 e^{-2\theta\tau}\big[\frac{\theta}{1-X} + \frac{X}{(1-X)^2}\big]$. Since $p_{-k} = \bar p_k$, P is real, $\sum_{k\ne0}\cos(qk)\operatorname{Re} p_k = \frac12(P(\theta) + P(1-\theta))$ and $\sum_{k\ne0}\sin(qk)\operatorname{Im} p_k = \frac12(P(1-\theta) - P(\theta))$. The bracket in P is the quantity W_m/N of Thm. 11 (with $\xi^m = e^{-2\theta\tau}$), so by its proof $\frac12(P(\theta)+P(1-\theta)) = \frac14 - \tau^2 Q/(4\sinh^2\tau)$ and $\frac12(P(1-\theta)-P(\theta)) = \tau^2\gamma$. Adding up: $M_{xx} = \frac{\tau^2}{12} - \frac14 + \frac{\tau^2}{4\sinh^2\tau} - \big(\frac{\tau^2}{12} - \frac{\tau^2\theta(1-\theta)}{2}\big) + \frac14 - \frac{\tau^2 Q}{4\sinh^2\tau} = \tau^2\beta$, $M_{yy} = \tau\coth\tau - \frac12 - \frac{\tau^2(1-x^2)}{8} - \frac{\tau^2(1+Q)}{4\sinh^2\tau} = \tau^2\alpha$ and $M_{xy} = i\tau^2\gamma$; finally $(\ell/2\pi)^2 = \tau^{-2}$. □

`large_n.out` (item 6) confirms the identity against direct lattice sums to $4\cdot10^{-12}$.

Corollary 20 (critical spacing of the row; PROVED)**.** *The horocyclic row with spacing ℓ (in curvature radii) is spectrally stable, i.e. every Bloch mode has purely imaginary exponents, if and only if $\ell \ge \ell_* = \pi/\eta_\infty = 0.8439\ldots$. The Bloch matrix $M(\theta)$ is positive definite for every $0 < \theta < 1$ if and only if $\ell > \ell_*$; at $\ell = \ell_*$ it degenerates exactly in the pairing mode $\theta = \frac12$. So the spectral and the definiteness thresholds coincide: there is no Krein window.*

Proof. The equations of motion are $\dot q = y^2 J_0 \nabla F$ in the co-moving frame. Since $dF = 0$ at the row, the variation of the Poisson tensor does not contribute, and the linearisation at $y = 1$ is $J_0 \operatorname{Hess} F$.

On a Bloch wave this is $J_0 M(\theta)$, so as in Prop. 3 the exponents are $\sigma = -i\gamma' \pm \sqrt{-M_{xx}M_{yy}}$ with $\gamma' = \tau^2\gamma$ real, and stability means $M_{xx}M_{yy} \geq 0$ for every θ. At $\theta = 0$, $\beta = \gamma = 0$ and $M(0) = \mathrm{diag}(0, \tau^2\alpha(1,\tau))$ with $\tau^2\alpha(1,\tau) = (\tau\coth\tau - 1) + \frac{1}{2}(1 - \tau^2/\sinh^2\tau) > 0$: both exponents vanish, and the null direction is the translation symmetry. For $0 < \theta < 1$, by Prop. 19, $M_{xx} \propto \beta > 0$ (Lemma 12) and $M_{yy} \propto \alpha = \Phi(\tau) + \psi(0,\tau) - \psi(x,\tau) \geq \Phi(\tau)$, with equality at $\theta = \frac{1}{2}$ (Lemma 13). So the row is stable iff $\Phi(2\pi/\ell) \geq 0$, i.e. iff $\ell \geq \ell_*$. For definiteness, $\alpha\beta - \gamma^2 = \beta(\Phi + G)$ with $G \geq 0$, $G = 0$ only at $x = 0$ (Lemma 17). □

Spectral stability here is not bounded linear stability: there is a Jordan block at $\theta = 0$ for every ℓ (a uniform change of height makes the row drift), and at $\theta = \frac{1}{2}$ when $\ell = \ell_*$, so those modes grow linearly. Positive definiteness also holds only mode by mode: the smallest eigenvalue of $M(\theta)$ tends to 0 as $\theta \to 0$ ($\beta = O(\theta^2)$), as for mode 0 of the ring, and we make no energy–momentum or nonlinear claim for the infinite row. A direct simulation of a 12-periodic row (finite differences of the full velocity field, all images summed) reproduces these exponents to $2 \cdot 10^{-7}$ and has growth rates 2.60, 0.98, 0.47 at $\ell/\ell_* = 0.9$, 0.98, 0.995 and none above ℓ_* (NUMERICAL; this simulation is not among the recorded outputs of the repository). So Theorem 16 says that a large ring loses stability exactly where the horocyclic row with the same spacing does. The critical mode $\theta = \frac{1}{2}$ displaces neighbours in opposite normal directions (its growing eigenvector also has a tangential part): it is the pairing instability of a vortex row.

Why curvature stabilises. In (5) the dense limit $\ell \ll R$ ($\eta = \pi/\ell \to \infty$) gives $8\Phi \to -1$, unstable at every spacing, as for a straight row in the plane (the classical row instability [7]). The sparse limit $\ell \gg R$ gives $8\Phi \to 2$, stable. Screening alone does not explain this. The tanh interaction is screened beyond one curvature radius ($-\ln\tanh(d/2) \approx 2e^{-d}$), but a row along a *geodesic* of $\mathbb{H}^2$, with the same interaction and the same screening, is unstable at every spacing (subsection "Rows on curves of constant geodesic curvature" at the end of this section). What stabilises the horocyclic row at large spacing is the bending of the horocycle itself, whose geodesic curvature is 1.

Odd N: the threshold is unique

Proposition 21 (PROVED)**.** *For every odd $N \geq 17$, $a_{(N-1)/2}$ has exactly one zero in $\xi \in (0,1)$, at which it changes sign from negative to positive; equivalently, as a function of τ it is positive on $(0,5]$, negative on $[10,\infty)$ and strictly decreasing on $[5,10]$. With Cor. 14, Prop. 15 (even N) and Thm. 7(b) (odd $N \leq 15$), every $N \geq 8$ has exactly one spectral threshold; by Thm. 7(a) every $N \leq 7$ is stable at every radius.*

Proof. Write $x = 1/N$, so $a_{(N-1)/2}/N^2 = \Phi_N(\tau) + \varepsilon_N(\tau)$ with $\Phi_N = \Phi + \Delta_N$ strictly decreasing (the expression in Prop. 15 equals $8\Phi_N$ for every N, and its monotonicity proof needs only $N/2 > 1$) and $\varepsilon_N = \psi(0,\tau) - \psi(x,\tau) = \frac{x^2}{8} + \frac{\tilde{D}}{4\sinh^2\tau}$, $\tilde{D} := \cosh\tau - Q(x,\tau)$.

(i) $0 \leq \varepsilon_N \leq x^2/4$. The lower bound is Lemma 13. For the upper bound let $D = \frac{x^2}{2}\sinh^2\tau - \tilde{D}$; then $D(0) = 0$ and $D' = \sinh\tau\,[x^2(\cosh\tau - \cosh x\tau) + (\cosh x\tau - 1)] \geq 0$, so $\tilde{D} \leq \frac{x^2}{2}\sinh^2\tau$. The lower bound $\tilde{D} \geq -\frac{x^2}{2}\sinh^2\tau$ is $\varepsilon_N \geq 0$. Hence $|\tilde{D}| \leq \frac{x^2}{2}\sinh^2\tau$ (and $\tilde{D}$ does take both signs).

(ii) Location. By the bounds on Δ_N in the proof of Thm. 16, $-\frac{1}{2N^2} - \frac{\delta^2}{2} \leq \Delta_N \leq \frac{\tau}{3N^2}$. With $\Phi(5) = 0.0516\ldots$ and $\Phi(10) = -0.0300\ldots$ this gives, for $N \geq 17$, $\Phi_N(5) > 0.049$ and $\Phi_N(10) + x^2/4 < -0.017$. As Φ_N decreases and $0 \leq \varepsilon_N \leq x^2/4$, the critical block is positive for $\tau \leq 5$ and negative for $\tau \geq 10$.

(iii) Slope on $[5,10]$. With $\eta=\tau/2$, $-\Phi'=\frac18\big[(\tanh\eta-\eta\operatorname{sech}^2\eta)/\eta^2+\mathcal{L}\mathcal{L}'\big]$. On $\eta\in[2.5,5]$: $\tanh\eta-\eta\operatorname{sech}^2\eta$ is increasing (its derivative is $2\eta\operatorname{sech}^2\eta\tanh\eta$), so it is at least its value 0.920 at 2.5; $\mathcal{L}\ge 1-1/\eta\ge 0.6$; and $\mathcal{L}'\ge\frac{1}{25}-\sinh^{-2}2.5>0.0126$. So $-\Phi'\ge\frac18(0.92/25+0.6\cdot0.0126)>0.0055$. Next, $\Delta_N'=\delta'(\coth\tau-\frac1\tau-\delta)+\delta(\frac{1}{\tau^2}-\frac{1}{\sinh^2\tau})$ with $0<\delta'=\mathcal{L}'(\tau/N)/N^2\le\frac{1}{3N^2}$, $0<\coth\tau-\frac1\tau-\delta<0.91$, and $0<\delta<\frac{\tau}{3N^2}$ with $\frac{1}{\tau^2}-\frac{1}{\sinh^2\tau}\le\frac{1}{\tau^2}$, so $|\Delta_N'|\le\frac{0.91}{3N^2}+\frac{1}{3\tau N^2}\le\big(\frac{0.91}{3}+\frac{1}{15}\big)\frac{1}{N^2}=\frac{0.37}{N^2}$. Finally $\tilde D'=\sinh\tau\,[1-(1-x^2)\cosh x\tau]$ and $|1-(1-x^2)\cosh w|\le\cosh w\,(x^2+w^2/2)$, so with (i)

$$|\varepsilon_N'|\le\frac{|\tilde D'|}{4\sinh^2\tau}+\frac{|\tilde D|\cosh\tau}{2\sinh^3\tau}\le x^2\Big[\frac{\cosh(x\tau)(1+\tau^2/2)}{4\sinh\tau}+\frac{\coth\tau}{4}\Big]\le 0.31\,x^2$$

on $[5,10]$ for $x\le\frac{1}{17}$, since $(1+\tau^2/2)/\sinh\tau$ is decreasing there, $\cosh(x\tau)\le\cosh\frac{10}{17}$ and $\coth\tau\le\coth 5$ (the bracket is at most 0.304). Altogether $\frac{d}{d\tau}(a_{\rm crit}/N^2)\le-0.0055+0.68/N^2<0$ for $N\ge17$. □

The constants are crude: on $[5,10]$ the true values are $-\Phi'\ge0.0089$, $|\Delta_N'|\le1/(3N^2)$ and $|\varepsilon_N'|\le0.012\,x^2$ (`large_n.out`, item 9). For $N\le15$ the certified Thm. 7 gives uniqueness.

Rows on curves of constant geodesic curvature

Status: for the tanh model the exact stability criterion (a one-variable series), the critical curvature $1/\sqrt2$ *and an explicit stability bound are DERIVED (by hand, sympy-checked, unrefereed); that the threshold is a single curve is CERTIFIED (Arb ball arithmetic); the values* $\ell^*(\kappa)$ *are computed; exploratory, scripts in* `explorations/2026-10-14/rows/`. The horocycle belongs to a one-parameter family of homogeneous curves in $\mathbb{H}^2$, distinguished by their geodesic curvature κ: geodesics ($\kappa=0$), hypercycles ($0<\kappa<1$, the curves at constant distance h from a geodesic, $\kappa=\tanh h$), horocycles ($\kappa=1$) and circles ($\kappa=\coth r>1$). Vortices equally spaced along any of them form a relative equilibrium; for circles these are the rings of the earlier sections. In the upper half-plane write $z=e^{u+i\varphi}$. The hypercycle $\varphi=\alpha$ has $\kappa=\cos\alpha$, the metric is $|du+i\,d\varphi|^2/\sin^2\varphi$, and

$$\cosh d_{ij}=1+\frac{\cosh(u_i-u_j)-\cos(\varphi_i-\varphi_j)}{\sin\varphi_i\sin\varphi_j},\qquad \dot u_i=\sin^2\varphi_i\,\partial_{\varphi_i}H,\qquad \dot\varphi_i=-\sin^2\varphi_i\,\partial_{u_i}H.$$

The row $u_k=k\ell'$, $\varphi_k=\alpha$ (arc spacing $\ell=\ell'/\sin\alpha$) translates along the hypercycle.

Proposition 22 (large-spacing criterion; DERIVED at leading order)**.** *Let* $W(d)\sim A\,e^{-\gamma d}$ *as* $d\to\infty$*, with* $A,\gamma>0$ *(the tanh model has* $A=2$*,* $\gamma=1$*). To leading order in* $e^{-\gamma\ell'}$*, the row on the hypercycle of curvature* κ *is spectrally stable if and only if* $\kappa\ge1/\sqrt2$*, whatever* A *and* γ*. The critical mode is the pairing mode* $\theta=\frac12$*.*

Derivation. At large spacing only nearest neighbours contribute at leading order, and $e^{-\gamma d}\simeq(\sin\varphi_i\sin\varphi_j)^\gamma e^{-\gamma|u_i-u_j|}$ up to a constant factor. Linearise at $u_k=k\ell'+U_k$, $\varphi_k=\alpha+P_k$ and take Bloch waves e^{iqk}. The 2×2 symbol has equal diagonal entries, and its off-diagonal entries are positive multiples of

$$\gamma\cos^2\alpha\,(1+\cos q)+2\cos^2\alpha-1\qquad\text{and}\qquad\cos q-1\le0.$$

So the exponents are purely imaginary for every q if and only if the first is ≥0 for every q. Its minimum, at $q=\pi$, is $\cos2\alpha$, which is ≥0 if and only if $\cos\alpha\ge1/\sqrt2$ (sympy: `asymptotic_check.py`). □

For the tanh model the conclusion holds exactly, at every spacing:

Exact symbol (tanh model; DERIVED, sympy-checked). Write $K = \kappa^2$, $c_j = \cosh(j\ell')$ and $D_j = c_j + 1 - 2K = c_j - \cos 2\alpha > 0$. Each pair contribution to $\dot{u}_0$ is even in $u_0 - u_j$ and each contribution to $\dot{\varphi}_0$ is odd. Hence at Bloch wavenumber q the symbol is $\left(\begin{smallmatrix} i\delta & B \\ C & i\delta \end{smallmatrix}\right)$, with equal, purely imaginary diagonal entries, $\delta(q) = -2\sqrt{K}\sin^3\alpha \sum_{j\geq 1} \sinh(j\ell')\sin(qj)/D_j^2$, and

$$B(q) = 2\sum_{j\geq 1}\big[a(c_j) + b(c_j)\cos qj\big], \qquad C(q) = 2\sum_{j\geq 1} d(c_j)\,(1-\cos qj),$$

$$a(c) = \sin^2\alpha\,\frac{(3K-1)c^2 - (4K^2-K+1)c + 2K^2}{(c-1)D^2},$$
$$b(c) = \sin^2\alpha\,\frac{Kc^2 + (1-3K)c + 2K^2 - 2K + 1}{(c-1)D^2}, \qquad d(c) = -\sin^4\alpha\,\frac{c^2+c-2K}{(c-1)D^2}$$

(`symbol_q.py`). Since $d < 0$, $C(q) < 0$ for $q \neq 0$, and the exponents are $i\delta \pm \sqrt{BC}$. So the row is spectrally stable if and only if

$$F(q) := \sum_{j\geq 1}\big[a(c_j) + b(c_j)\cos qj\big] \;\geq\; 0 \qquad \text{for all } q.$$

This reproduces the supercell growth rates to 7 digits and the thresholds $\ell^*(\kappa)$ below to 6–7 digits, up to $\kappa = 0.999$ (`exact_symbol.py`). The minimum over q is always at $q = \pi$; see (d) below. Two identities do the work: $b(c) > 0$ for $c > 1$ (in $t = c - 1$ its numerator has coefficients K, $1-K$, $2(1-K)^2$), and

$$a - b = \sin^2\alpha\,\frac{(2K-1)c - 1}{(c-1)D}, \qquad a + b = \sin^2\alpha\,\frac{(4K-1)c - (4K^2-2K+1)}{D^2}.$$

Consequences (tanh model; DERIVED, sympy-checked). (a) *Stability.* If $\kappa > 1/\sqrt{2}$ and $\cosh\ell' \geq 1/(2\kappa^2-1)$, then $a_j - b_j \geq 0$ for every j (strictly for $j \geq 2$), so $F(q) \geq \sum_j (a_j - b_j) > 0$ and the row is stable. In arc length the condition is $\ell \geq \ell_A(\kappa) = \operatorname{arcosh}\big(1/(2\kappa^2-1)\big)/\sqrt{1-\kappa^2}$, which exceeds the true threshold by a factor 1.2–2.3 (dashed in Fig. 1) and tends to 2 as $\kappa \to 1$ (`stability_bound.py`). (b) *All-mode instability for $\kappa \leq \frac{1}{2}$.* Then $4K - 1 \leq 0$, so $a_j + b_j < 0$ for every j and $F(q) \leq \sum_j (a_j + b_j) < 0$ for every q: every Bloch mode $q \neq 0$ grows, at every spacing. (c) *Instability for $\kappa \leq 1/\sqrt{2}$ at every spacing.* At $q = \pi$, $F(\pi) = \sin^2\alpha \sum_{j\geq 1} g_j$ with $g_j = (a_j - b_j)/\sin^2\alpha$ for odd j and $(a_j + b_j)/\sin^2\alpha$ for even j, i.e.

$$g_j = \frac{(2K-1)c_j - 1}{(c_j - 1)D_j}\ (j \text{ odd}), \qquad g_j = \frac{(4K-1)c_j - (4K^2-2K+1)}{D_j^2}\ (j \text{ even}).$$

Every g_j is strictly increasing in K ($\partial_K g = 2(c+1)/D^2$ for odd j and $2(2c^2+c-1-2K)/D^3$ for even j), so for $K \leq \frac{1}{2}$ the sum is at most its value at $K = \frac{1}{2}$,

$$G(\ell') = \sum_{m\geq 1}\Big[-\frac{1}{(c_{2m-1}-1)c_{2m-1}} + \frac{c_{2m}-1}{c_{2m}^2}\Big],$$

and $G < 0$ for every $\ell' > 0$. For $\ell' \geq 0.75$ keep only $j = 1$ among the odd terms and use $(c-1)/c^2 < \operatorname{sech}(j\ell') < 2e^{-j\ell'}$: with $q_0 = e^{-2\ell'}$, $G < -4q_0/(1+q_0)^2 + 2q_0/(1-q_0)$, which is negative for $q_0 < \sqrt{5} - 2$, i.e. $\ell' > 0.722$. For $\ell' \leq 1.2$ bound the even terms by $\int_0^\infty \operatorname{sech}(2\ell' x)\,dx = \pi/(4\ell')$;

then $G<0$ follows from $(\cosh x-1)\cosh x<4x/\pi$ on $(0,1.2]$, which holds because the difference is convex, vanishes at 0 and is -0.060 at 1.2. So $B(\pi)<0<-C(\pi)$ and the pairing mode grows at the real rate $\sqrt{B(\pi)C(\pi)}$ (`instability_proof_check.py`). The even terms alone are positive at large c_j when $\frac14<K<\frac12$ (`pair_sign.py`); only the sum is signed.

(d) *The critical mode is the pairing mode, and the exact threshold.* With $z=e^{-x}$ and $c=\cosh x$, the pair functions expand as

$$b=2\sin^2\alpha\sum_{n\ge1}n\cos^2(n\alpha)\,z^n,\qquad d=-2\sin^2\alpha\sum_{n\ge1}n\sin^2(n\alpha)\,z^n,\qquad a=d+\frac{\sin^2 2\alpha}{2(c-\cos 2\alpha)}$$

(equivalently $b-d=\sin^2\alpha/(c-1)$; exact identities, `qpi_proof.py`). All coefficients of b are ≥0, so with $r_n=e^{-n\ell'}$,

$$\sum_{j\ge1}b(c_j)\cos qj=2\sin^2\alpha\sum_{n\ge1}n\cos^2(n\alpha)\,P(r_n,q),\qquad P(r,q)=\sum_{j\ge1}r^j\cos qj=\frac{r\cos q-r^2}{1-2r\cos q+r^2},$$

and $\partial_qP=-r(1-r^2)\sin q/(1-2r\cos q+r^2)^2<0$ on $(0,\pi)$. So F is strictly decreasing on $(0,\pi)$. The row is stable if and only if $F(\pi)\ge0$, and a necklace of N vortices (allowed $q=2\pi m/N$) is stable if and only if $F(2\pi\lfloor N/2\rfloor/N)\ge0$. Summing the geometric series gives the threshold function as one rapidly convergent series,

$$F(\pi)=\sum_{n\ge1}\Big[\big(\sin 2\alpha\sin 2n\alpha-2n\sin^2\alpha\sin^2 n\alpha\big)\frac{r_n}{1-r_n}-2n\sin^2\alpha\cos^2 n\alpha\,\frac{r_n}{1+r_n}\Big],$$

whose leading term at large spacing is $2\sin^2\alpha\cos 2\alpha\,e^{-\ell'}$. Its root reproduces the supercell thresholds (to 10^{-6}), and for every $\kappa\in(1/\sqrt2,1)$ it changes sign exactly once in ℓ' (CERTIFIED, (g) below); e.g. $\ell^*(0.8)=2.2554766253$. As $\alpha\to0$ at fixed arc spacing $\ell=\ell'/\sin\alpha$, $n\alpha\to t$ and the series becomes

$$\int_0^\infty\Big[\frac{2\sin 2t-2t\sin^2 t}{e^{\ell t}-1}-\frac{2t\cos^2 t}{e^{\ell t}+1}\Big]dt=-\tfrac14\eta^2\coth^2\eta+\eta\coth 2\eta-\tfrac14,\qquad\eta=\pi/\ell.$$

This vanishes exactly when $(\coth\eta-1/\eta)^2=2\tanh\eta/\eta$, the horocycle criterion of Cor. 20 (`horo_limit.py`). So the hypercycle family limits onto the horocycle row, with the same ℓ_*.

(e) *Why this works, and other kernels.* In the coordinates (u,φ) the operator $-\Delta+\nu(\nu+1)$ becomes $\sin^2\varphi\,[-\partial_u^2-\partial_\varphi^2+\nu(\nu+1)/\sin^2\varphi]$, which separates. The Pöschl–Teller modes on $(0,\pi)$ are $\psi_n=\sin^{\nu+1}\varphi\,C^{(\nu+1)}_{n-\nu-1}(\cos\varphi)$, with eigenvalue n^2 and $n\ge\nu+1$. So the kernel $W_\nu=2\pi G_\nu$ ($\nu=0$: the tanh model; $\nu=1$: $Q_1(\cosh d)$) is $W_\nu=2\pi\sum_n e^{-n|u-u_0|}\psi_n(\varphi)\psi_n(\varphi_0)/(2nN_n)$, $N_n=\int_0^\pi\psi_n^2$. With $w_n(x)=2\pi e^{-nx}/(2nN_n)$,

$$b=\sin^2\alpha\sum_n w_n\,\psi_n'(\alpha)^2\ \ge0,\qquad d=-\sin^2\alpha\sum_n n^2w_n\,\psi_n(\alpha)^2\ \le0,$$
$$a=\sum_n w_n\big[2\sin\alpha\cos\alpha\,\psi_n\psi_n'+(\nu(\nu+1)-n^2\sin^2\alpha)\psi_n^2\big].$$

For $\nu=0$ ($\psi_n=\sin n\varphi$) these are the expansions in (d). So for every $\nu\ge0$ the symbol has the same structure, the critical mode is $q=\pi$, and the threshold is the zero of the analogous one-variable series $F_\nu(\pi)$ (`modes_general.py`; checked against direct lattice sums to 10^{-11}). The lowest mode is $\psi\propto\sin^m\varphi$, $m=\nu+1$, and its bracket in $F_\nu(\pi)$ equals $m\sin^{2m}\alpha\cos 2\alpha$ exactly.

So the leading large-spacing term is $\propto \cos 2\alpha\, e^{-m\ell'}$ for every ν, which is the content of Prop. 22 for this family. The reason ν drops out: in proper coordinates along the curve the bracket is $\kappa\,\partial_\rho(\psi^2) - P$, where $P = (\partial_\rho\psi)^2 + (\partial_\tau\psi)^2 - \nu(\nu+1)\psi^2$ and the lowest mode has frequency m. For $\psi = \sin^m\varphi$ one finds $P = m\psi^2$ and $\kappa\,\partial_\rho(\psi^2) = 2m\kappa^2\psi^2$, so both terms scale with m (`explorations/2026-10-14/lorentzian/ground_mode.py`). Geometrically, the hypercycle $\varphi = \alpha$ meets the ideal boundary at angle α, so the critical hypercycle is the one meeting it at 45°. For $\nu = 1$ the roots of $F_1(\pi)$ reproduce the supercell thresholds ($\ell^* = 4.0856$, 3.0055, 1.9820 at $\kappa = 0.75$, 0.8, 0.9). As $\kappa \to 1$ they tend to 1.4491, the critical spacing of the horocyclic row for this kernel, which is the zero of the $\alpha \to 0$ integral. For $\nu = 1, 2$, $F_\nu(\pi) < 0$ at every computed point with $\kappa \le 1/\sqrt{2}$, and instability at every spacing is proved for $\nu = 0$ in (c) and CERTIFIED for $\nu = 1$ (Arb; `explorations/2026-10-14/rows/q1_instability_cert.py`): for $\ell' \ge 4.5$ an explicit expansion $F_1 = \frac{8}{3}\sin^4\alpha\cos 2\alpha\, r^2 + c_3 r^3 + O(r^4)$ with $c_3 \le -\frac{4}{3}$; for $0.05 \le \ell' \le 4.5$, 6781 boxes; for $\ell' \le 0.05$, elementary bounds from the positivity and monotonicity of the mode expansions. So $\kappa_c = 1/\sqrt{2}$ is exact for $\Delta = 2$ as well.

(f) *Closed form.* Since $r/(1+r) = r/(1-r) - 2r^2/(1-r^2)$, the series for $F(\pi)$ collapses to Jacobi theta functions with nomes $e^{-\ell'/2}$ and $e^{-\ell'}$, plus the Eisenstein sum $E(r) = \sum nr^n/(1-r^n)$. The modular transformation $\tau \mapsto -1/\tau$ then gives an exact dual series. With $\eta = \pi\alpha/\ell'$, $Q_1 = e^{-4\pi^2/\ell'}$ and $Q_2 = e^{-2\pi^2/\ell'}$,

$$F(\pi) = F_0 - \frac{2\pi}{\ell'}\sum_{n\ge1}\frac{Q_1^n}{1-Q_1^n}\Big[\sin 2\alpha\sinh\frac{4\pi n\alpha}{\ell'} - \frac{4\pi n\sin^2\alpha}{\ell'}\Big] - \frac{2\pi^2\sin^2\alpha}{\ell'^2}\sum_{n\ge1}\frac{nQ_2^n}{1-Q_2^n}\Big[1+\cosh\frac{2\pi n\alpha}{\ell'}\Big],$$

$$F_0 = \frac{\pi\sin 2\alpha}{2\ell'}\coth 2\eta - \frac{\pi^2\sin^2\alpha}{4\ell'^2}\coth^2\eta - \frac{\alpha\sin 2\alpha}{\ell'} - \frac{\cos 2\alpha}{4}$$

(`dual_series.py`; the three forms agree to 10^{-36}). The corrections are $O(e^{-2\pi(\pi-\alpha)/\ell'})$. For $\kappa \ge 1/\sqrt{2}$ both sums are positive, since $\sinh x \ge x$ and $\alpha\sin 2\alpha - \sin^2\alpha = \sin\alpha\,(2\alpha\cos\alpha - \sin\alpha) > 0$, so $F(\pi) < F_0$ there. So the elementary equation $F_0 = 0$ is a closed-form threshold. Its root agrees with $\ell^*(\kappa)$ to 10^{-4} at $\kappa = 0.8$, 10^{-11} at 0.9 and 10^{-20} at 0.95, and wherever $F_0 < 0$ the row is unstable (Fig. 2). As $\alpha \to 0$ at fixed arc spacing, F_0 becomes the horocycle criterion of (d) exactly. Near $\kappa = 1/\sqrt{2}$, where ℓ' is large, the original series is the useful one.

(g) *The threshold is a single curve* (CERTIFIED; `single_root_cert.py`, Arb ball arithmetic via python-flint, a few seconds). For every $\kappa \in (1/\sqrt{2}, 1)$, $F(\pi)$ has exactly one root $\ell'^*(\kappa)$, and it is transversal: the row is unstable for $\ell' < \ell'^*$ and stable for $\ell' > \ell'^*$. It suffices to show $\partial_{\ell'}F > 0$ at every zero with $\alpha < \pi/4$. Then $F(\alpha, \cdot)$ can only cross zero upwards, and it does cross, since $F < 0$ at small spacing and $F > 0$ by (a). Zeros with $\alpha \ge \pi/4$ are excluded by (c). The zeros are covered by three regions.

- $\ell' \ge 6$: here $G = F/\sin^2\alpha = P_1 r + P_2 r^2 + E$, with $r = e^{-\ell'}$, $P_1 = 2\cos 2\alpha$ and $|E| + |rE_r| \le 0.32\, r^2$. At a zero this gives $\partial_{\ell'}G = -P_2 r^2 + E - rE_r$. By (a) every zero has $2\kappa^2 - 1 < 1/\cosh 6$, so there $-P_2 \in [1.95, 2.01]$.

- $0.25 \le \ell' \le 6$: 15,225 adaptive boxes. On each box the Lambert series of G, with tail bounds, either excludes 0 or gives $\partial_{\ell'}G > 0$.

- $\ell' < 0.25$, in the variables (α, η) with $\eta = \pi\alpha/\ell'$ and the dual series $F = F_0 - R$, $0 \le R$: for $\eta \ge 7$, $F \le F_0 < 0$ by an elementary estimate; for $\eta \le 2.5$, $F_0 \ge h(\eta) - \eta^2\big[\frac{2t^2}{3}(\frac{1}{2}+\eta) + \frac{2t}{\pi}\big]$

with $t = 0.25/\pi$ and h the horocycle function, which exceeds R (margin $0.022\,\eta^2$ against $R \le 3\cdot 10^{-32}\eta^2$); in between, 113 boxes, each with $F_0 - R_{\max} > 0$, $F_0 < 0$, or $\partial_\eta F_0 + \max|\partial_\eta R| < 0$, using $\partial_{\ell'} F|_\alpha = -(\eta/\ell')\partial_\eta F$.

So the stable set of hypercycle rows is exactly $\{\ell \ge \ell^*(\kappa)\}$, with ℓ^* real-analytic (by transversality). Since every g_j in (c) increases with κ^2, $\ell'^*(\kappa)$ is strictly decreasing. By (a), $\ell'^* \to \infty$ as $\kappa \downarrow 1/\sqrt{2}$. In arc length the computed $\ell^*(\kappa)$ decreases to the horocyclic value 0.8439.

By (a) and (c), $1/\sqrt{2}$ is the exact critical curvature of the tanh model: at and below it no spacing is stable, and above it every spacing beyond $\ell_A(\kappa)$ is. The kernel-independence in Prop. 22 remains a leading-order statement.

Numerical results (tanh model). The full linearisation (supercells of 40 vortices with periodic images, `rows.py`) gives the following. (i) Rows on geodesics and on the hypercycle $\kappa = 0.5$ are unstable at every spacing tested, $0.5 \le \ell \le 80$. (ii) At spacings 40 and 80 the critical curvature is 0.7071064. (iii) The critical spacing $\ell^*(\kappa)$ is 4.76, 3.19, 2.26, 1.44, 1.01, 0.874 at $\kappa = 0.72$, 0.75, 0.8, 0.9, 0.98, 0.999; it diverges at $\kappa = 1/\sqrt{2}$ and approaches the horocyclic value $\ell_* = 0.8439$ as $\kappa \to 1$. (iv) Just below threshold the growing mode is always $\theta = \frac{1}{2}$, and the threshold is the zero of a single lattice sum, the $\theta = \frac{1}{2}$ off-diagonal entry $A_{UP}(\pi) = B(\pi)$ (to 6–7 digits with the closed forms above). (v) For the screened kernel $W = Q_1(\cosh d)$ (the Green's function of $\Delta - 2$, $\gamma = 2$) the critical curvature is again 0.7071061, while $\ell^*(\kappa)$ is larger (1.98 at $\kappa = 0.9$). With $\epsilon = e^{-\ell'}$ the pairing sum of the tanh model expands as

$$\frac{A_{UP}(\pi)}{4\sin^2\alpha} = (2\kappa^2 - 1)\,\epsilon + (8\kappa^4 - 3)\,\epsilon^2 + 4(8\kappa^6 - 8\kappa^4 + 2\kappa^2 - 1)\,\epsilon^3 + O(\epsilon^4)$$

(sympy, `next_order.py`, checked against the exact sum), so $\ell^*(\kappa) \simeq -\ln(2\kappa^2 - 1)/\sqrt{1 - \kappa^2}$ as $\kappa \downarrow 1/\sqrt{2}$, within 0.1% at $\kappa = 0.72$. The series is only asymptotic.

For $\kappa > 1$ the same boundary is traced by the rings. The threshold ring of N vortices has $\kappa = \coth r_*$ and spacing $\ell_N = 2\pi \sinh r_*/N$, and Thm. 16 gives $(\kappa, \ell_N) \to (1, \ell_*)$. So circles, the horocycle and hypercycles carry one continuous stability boundary (Fig. 1).

Remark (closed surfaces). Near a closed geodesic of length L a hyperbolic surface is the cylinder $\mathbb{H}^2/\langle u \mapsto u + L\rangle$, and a necklace of N vortices at distance h from the geodesic is a hypercycle row with period L. By the collar lemma the collar has half-width $w = \operatorname{arsinh}(1/\sinh(L/2))$, so curvatures inside it satisfy $\kappa \le \tanh w$. For the systole of the Bolza surface ($\cosh(L/2) = 1 + \sqrt{2}$) this bound is exactly $\sqrt{2} - 1 < \frac{1}{2}$, so by Consequence (b) every Bloch mode of every necklace inside the collar grows: no necklace of $N \ge 2$ vortices, of either parity, is stable there (DERIVED). By (d) the necklace is stable exactly when $F(2\pi\lfloor N/2\rfloor/N) \ge 0$ at $\ell' = L/N$, $\kappa = \tanh h$; this gives the thresholds $h^* = 1.0574$, 1.1494, …, 2.1070 for $N = 2, \dots, 12$, as `collar.py` found by direct computation. The explicit bound (a) gives $\tanh^2 h \ge \frac{1}{2}\big(1 + 1/\cosh(L/N)\big)$, e.g. $h \ge 1.22$ for $N = 2$. Both statements are for the cylinder model only: the rest of the surface and the zero-net-circulation constraint are not modelled.

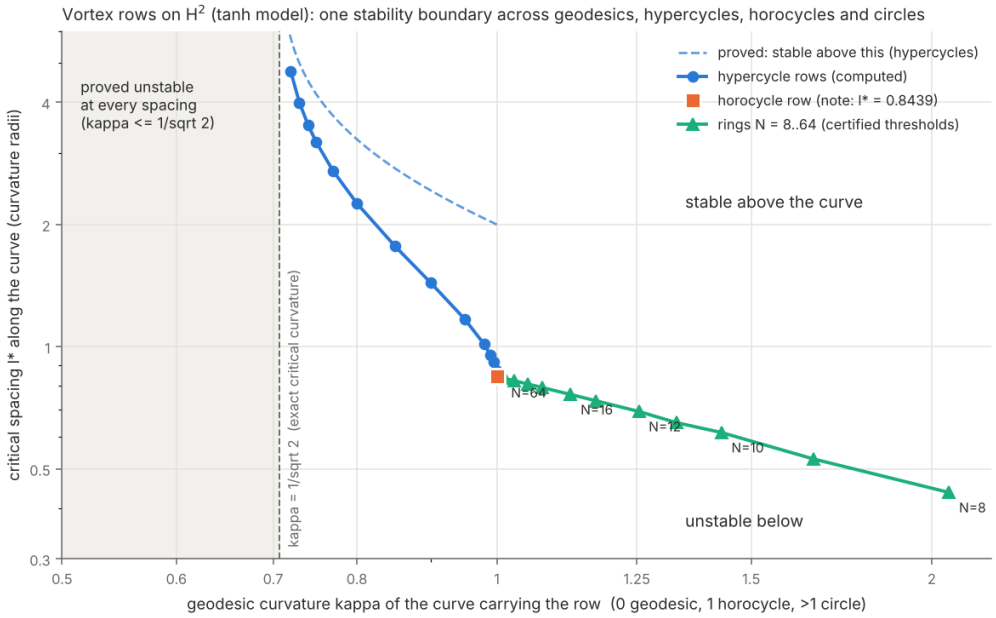


Figure 1: Critical spacing of vortex rows on curves of geodesic curvature κ in $\mathbb{H}^2$ (tanh model). Hypercycle branch: roots of the exact criterion $F(\pi) = 0$; dashed, the proved sufficient bound $\ell_A(\kappa)$. Horocycle point: Cor. 20. Ring branch: the certified thresholds for $N = 8$–64. At and below $\kappa = 1/\sqrt{2}$ no spacing is stable (proved).

10 Comparison with the sinh model and the flat disk

	sinh model [1]	tanh model (this note)
pair term	$-\ln\sinh(d/2)$	$-\ln\tanh(d/2)$
radial stiffness	$(N-1)(1+c^2)/2$	$(N-1)c - \frac{1}{2}(c-1)^2 + O(\xi^N)$
$s\theta$ coupling	none	$g_m = \frac{N}{2}(W_m - W_{N-m})$
threshold	$\cosh^2 r_* = q(N) \in \mathbb{Q}$	algebraic, degree (4)
$N \le 7$	stable for all r	stable for all r (cert.)
$\xi_*(8)$	$8 - 3\sqrt{7}$ (unit, deg 2)	unit, deg 10
large N	$\cosh r_* \sim \sqrt{N}/2$	$\cosh r_* \sim \kappa_\infty N$ (Thm. 16)
threshold spacing	$\to 0$	$\to \ell_* = 0.8439$
odd-N Krein window	none	width $\sim 10^{-7}$ ($N \le 15$), $\approx 0.0037/N^3$ (large N)

For $8 \le N \le 16$ the tanh threshold is larger:

N	8	9	10	11	12	13	14	15	16
$r_*^{\sinh}$	0.5119	0.6585	0.7954	0.8814	0.9692	1.0317	1.0967	1.1462	1.1978
$r_*^{\tanh}$	0.5312	0.6991	0.8654	0.9752	1.0917	1.1776	1.2691	1.3407	1.4168

Both radii grow logarithmically, so $r_*^{\tanh}/r_*^{\sinh} \to 2$, but slowly. This is consistent with the data to $N = 800$ (NUMERICAL).

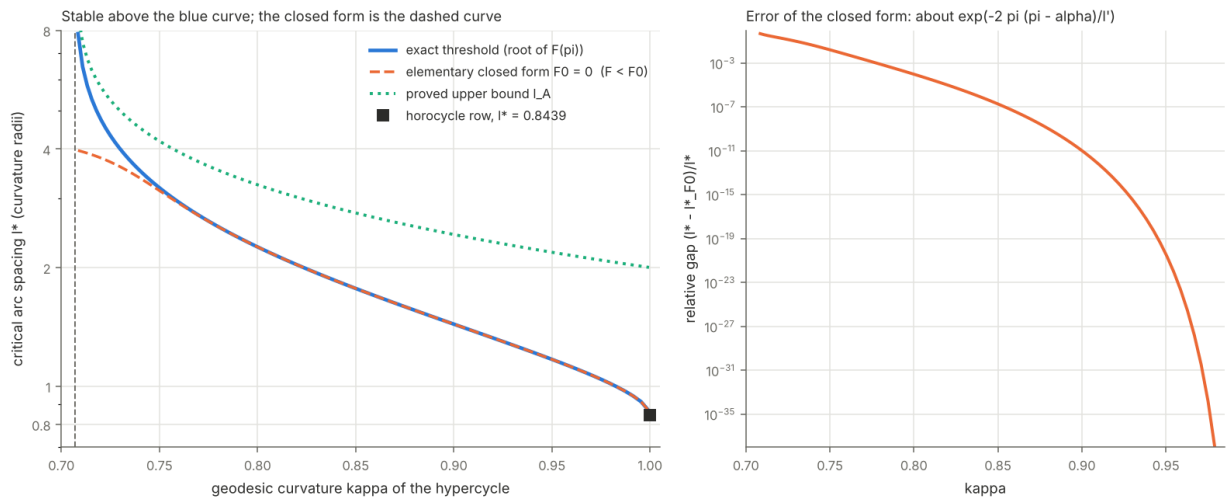


Figure 2: Left: the exact threshold of hypercycle rows (root of $F(\pi)$), the elementary closed form $F_0 = 0$ of (f), and the proved upper bound $\ell_A(\kappa)$. Right: relative gap between the exact and closed-form thresholds.

Interpretation. The models differ by the pair term $\ln\cosh(d/2)$, whose Laplacian is constant. In the sinh model this background gives a radial stiffness quadratic in $\cosh r$, which beats the Thomson destabilisation $N^2/8$ once $\cosh r \sim \sqrt{N}/2$. In the tanh model the interaction is screened by curvature: it decays like e^{-d} beyond one curvature radius. The stiffness is then only linear in $\cosh r$, up to exponentially small image terms, so stabilisation needs $\cosh r \sim 0.134N$.

Flat hard-wall disk (context only). This model has the Euclidean area form, image vortices, and the non-constant Robin self-image term (ring stability: Havelock [8]). It is not treated here. The point is only that the same image Green's function, with a different area form and a non-constant Robin function, is a different stability problem.

11 Limitations

- **Literature.** Boatto [2] computed the stability ranges of this model numerically, with the same block structure (§1). An implementation of her equations reproduces our values.
- **No human specialist review.** The derivations are checked by computer (exact computer algebra, Lean, and independent numerical recomputation). No human expert in point-vortex dynamics has reviewed them.
- **Large N.** The limit $\cosh r_*/N \to \kappa_\infty$ and the critical spacing are proved (Thm. 16), and so is the horocyclic-row picture (Prop. 19); the $1/N$ coefficients are derived without remainder bounds.
- **Rows on hypercycles.** For the tanh model the critical curvature $1/\sqrt{2}$, instability at and below it at every spacing, the sufficient bound $\ell_A(\kappa)$, the criterion $F(\pi) \geq 0$ and its horocycle limit are derived by hand with sympy checks (unrefereed); that $F(\pi)$ has a single root in ℓ' is certified with Arb (g), not proved on paper. For the screened kernels Q_ν the mode expansion (d)–(e) gives the same criterion and critical mode; instability below $1/\sqrt{2}$ is certified for $\nu = 1$

and numerical for $\nu \geq 2$. For kernels outside the family Q_ν (Prop. 22) only the leading-order criterion is derived.

- **Finite range.** Palindromy, irreducibility and the degree pattern are certified only for $N \leq 16$ (Conjecture 9). Irreducibility for $N \leq 16$ is certified by PARI/GP and checked in Lean.
- **Nonlinear stability.** This is stated only via Hessian definiteness and modulo $SO(2,1)$ (Remark 4). In the odd-N Krein window the ring is linearly (spectrally) stable and nonlinear stability is undecided; at the threshold itself it is open.

Files

`tanh/`: `blocks.py`, `exact.py`, `check_jacobian.py/.out`, `identities.py/.out`, `certify_v2.py/.out` (certification), `asymptotics.py/.out`, `large_n.py/.out` (§9), `irreducibility/` (PARI/GP irreducibility certificates: `extract.py/.out`, `irreducibility.gp/.out`, `polys.gp`), `boatto_check/` (Boatto's equations), and `review/` (independent Cartesian computations and symbolic checks).

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